Joint Code-Dimension and k-Galois-Hull Enumerators for Repeated-Root Constacyclic Codes over Square-Free Affine Algebras

Arun Kumar Yadav, Vipin Gupta, and Manju Pruthi

Abstract

We determine the exact joint distribution of the code dimension and the k-Galois hull dimension over all repeated-root constacyclic codes of a fixed length over a square-free affine algebra over a finite field. With length $n = n'p^t$ and component moduli that are p^t-th powers of square-free kernels, a code is encoded by one exponent at each distinct kernel factor, and the k-Galois dual acts on those factors through a Frobenius-twisted reciprocal permutation. On each orbit of this permutation the hull dimension is a cyclic sum of pairwise minima of consecutive exponents, so its bivariate law is the trace of a power of an explicit weighted transfer matrix with $p^t + 1$ states; multiplying over orbits, with a dimension-only factor for components whose twist is incompatible, gives a bivariate polynomial whose coefficients are the exact multiplicities of the pairs (code dimension, hull dimension). We obtain the marginal laws, the attainable pairs, closed means and variances, conditional laws, and the numbers of LCD, self-orthogonal, dual-containing and self-dual codes, and we give an algorithm that is polynomial in the length, the multiplicity and the number of kernel factors, checked against finite-field computations in characteristics 2 and 3.

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Supplementary material: the Python implementation of the enumerator and the complete verification log described in Section VI-C are included with the submission.

Index Terms

Constacyclic codes, Galois hulls, LCD codes, repeated-root codes, transfer matrices.

I. Introduction

The hull of a linear code C is the intersection of C with its dual. Codes with trivial hull are the linear complementary dual (LCD) codes introduced by Massey [1], which meet the Gilbert–Varshamov bound [2]; for cyclic codes a simple root condition characterizes them [3]. The hull dimension also enters exact counts of linear codes [4], and in the quantum setting it determines the number of maximally entangled pairs required by entanglement-assisted quantum codes [5], [6], a construction developed for cyclic and constacyclic families in [7]. The k-Galois hull, introduced by Fan and Zhang [8], replaces one slot of the coordinate pairing by a Frobenius power, unifying, at the field level, the Euclidean and Hermitian cases; Galois self-orthogonal constacyclic codes and MDS codes with Galois hulls of prescribed dimension are treated in [9] and [10]. For a constacyclic code, whose codewords are represented by polynomials modulo $x^n - \lambda$, duals and hulls are again ideals under suitable conditions, so factorization arguments apply [11].

Most Galois-hull results describe one code at a time. Liu and Pan [12] compute Galois hull dimensions of general linear codes from generator matrices. For constacyclic codes over finite fields, Debnath et al. [13] (see also the correction [14]) give a factor-theoretic hull-dimension formula and, under restrictions on the field parameters, count the codes with a prescribed hull dimension, extending the Euclidean and Hermitian enumeration of Sangwisut et al. [15]. At the level of generator descriptions, Taki Eldin and Solé [16] obtain generator polynomial matrices for Galois hulls of multi-twisted codes over finite fields, and intersections of linear and multi-twisted codes are described in [17]. A parallel line studies averages rather than full distributions: Skersys [18] for cyclic codes over fields, Jitman and Sangwisut [19] for Hermitian hulls, and Debnath and Prakash [20] for Galois hulls over finite fields and over a finite non-chain ring; the attainability of small Galois hull dimensions is studied in [21], and the sequence of numbers of linear codes with increasing hull dimension in [22]. Closest to the algebraic setting of this paper, Debnath, Islam, Martínez-Moro and Prakash [23] study k-Galois hulls of constacyclic codes over affine algebra rings: they describe dual and hull generators in idempotent form, derive a hull-dimension formula for an individual code under the hypotheses $\gcd(\operatorname{ord}\lambda, n') = 1$ and $\operatorname{ord}\lambda \mid 1 + p^{e-k}$ and give an LCD criterion and quantum-code applications. In the repeated-root regime $p \mid n$ the per-code hull-dimension

formula is also known for constacyclic codes over a single field [13], and we use it as a check rather than a claim (Remark 25). None of these works records how the pairs (code dimension, hull dimension) are distributed over a whole family.

The present paper is concerned with *repeated-root* lengths, that is, with $p \mid n$ in characteristic p, where the modulus $x^n - \lambda$ is a perfect power rather than a square-free polynomial. Repeated-root cyclic codes were investigated in generality by Castagnoli, Massey, Schoeller and von Seemann [24] and by van Lint [25], who showed that such families are asymptotically bad; nevertheless their finite algebraic structure generators, duals and distances at specific lengths has been developed extensively, for example in [26], in the study of constacyclic codes of prime-power length over rings [27], and in the chain-ring theory of [28]; see also [29]. Within the hull theme, Pang, Zhu and Li [30] characterize and enumerate the LCD repeated-root cyclic codes over a finite field, hulls of cyclic codes over $\mathbb{Z}_4$ are computed in [31], and Fan and Zhang [8] treat Galois self-dual constacyclic codes over fields, including repeated-root examples in characteristic 2. What is missing is the population-level picture for repeated-root lengths: the number of codes with each code dimension and each k-Galois hull dimension.

For a repeated-root length, write $n = n'p^t$ with $\gcd(n', p) = 1$ and put $M = p^t$. Componentwise, the Frobenius identity writes the modulus as the M-th power of a square-free kernel. A code is therefore determined by an exponent in $\{0, \dots, M\}$ at each distinct irreducible factor of that kernel. Along a compatible Frobenius-twisted reciprocal orbit, the hull statistic is a cyclic sum of pairwise minima of these exponents (Proposition 24), and its bivariate law is encoded by an $(M+1)$-state transfer matrix. The resulting formulas separate the variance into three orbit-length regimes; length-1 orbits can contribute positive hull dimension; and, for $t \geq 1$ in characteristic 2, self-dual codes can arise through the midpoint exponent $M/2$ (Section V).

We keep the coefficient algebra square-free, so that $\mathcal{A} \cong \prod_s K_s$ with $K_s = \mathbb{F}_{p^{d_s}}$, and allow the length to be arbitrary. All $\mathcal{A}$-linear λ-constacyclic codes of length $n = n'p^t$ are counted, as labeled submodules determined by the exponents of the distinct irreducible factors of the square-free kernel $x^{n'} - \lambda_0$, and all twists $\lambda \in \mathcal{A}^*$ are allowed: components whose twist is incompatible with the k-Galois dual contribute hull dimension 0 and are included with an explicit dimension-only factor. The main result (Theorem 28) is the exact joint enumerator

$$\begin{aligned} \mathscr{E}^{(M)}(U, Z) &= \sum_{C} U^{\kappa(C)} Z^{\eta_k(C)} \\ &= \prod_{s \text{ compatible}} \prod_{O \in \mathcal{O}_s} \operatorname{tr}\big((T_{w_O}^{(M)})^{a_O}\big) \end{aligned}$$

$$\times \prod_{s \text{ incompatible}} \prod_{f \in \mathcal{F}_s} (1 + U^{w_f} + \cdots + U^{M w_f}).$$

where the $(M+1) \times (M+1)$ transfer matrix $T_w^{(M)}$ has entries $U^{w(M-\beta)} Z^{w \min\{\alpha, M-\beta\}}$. On the overlap with [23], our hull-dimension formula agrees with theirs through the identity $M - \max\{u, v\} = \min\{M-u, M-v\}$ (Remark 25); the contribution of the present paper is the joint distribution and its consequences marginals; the multiplicity-independence of the LCD count in the fully compatible case; the exact support of attainable (code dimension, hull dimension) pairs; conditional laws and moments; extremal-class counts; and, given the orbit data, an exact algorithm polynomial in n, M and the number of distinct factors together with the removal, in this enumerative setting, of the coprimality hypothesis $\gcd(\operatorname{ord} \lambda, n') = 1$.

The argument follows the short pipeline in Figure 1. The Frobenius identity reduces each repeated-root modulus to the M-th power of a square-free kernel, so a code is recorded by one exponent at each distinct kernel factor. On each compatible reciprocal orbit, the hull dimension is a cyclic sum of pairwise minima whose bivariate law is a transfer-matrix trace; convolution over all compatible orbits, together with the dimension-only factors for incompatible components, gives the global enumerator. The support, moments, conditional laws, extremal-class counts, algorithm and examples are then obtained from this single generating function. Throughout, codes are counted as labeled $\mathcal{A}$-submodules, not up to monomial equivalence, and no claim is made about minimum distances. The paper is organized as follows. Section II introduces the repeated-root setting, the Frobenius reduction to the square-free kernel and the exponent-selection description of codes. Section III develops k-Galois duality, the compatibility criterion, reciprocal orbits and the incompatible case. Section IV proves the min-edge hull formula and the main joint-enumerator theorem, together with its marginal and symmetry consequences. Section V derives the LCD count, exact support, moments and conditional laws, and the self-orthogonal, dual-containing and self-dual counts. Section VI presents the computation of orbit data, the exact algorithm and its complexity, worked examples in characteristics 2, 3 and 5, and independent verification. Section VII summarizes the scope of the results and records open directions.

II. Affine algebras and repeated-root constacyclic codes

The coefficient algebra in this paper is a square-free affine algebra, whereas the code modulus is allowed to be non-square-free: we do not assume that $\gcd(n, p) = 1$, so $x^n - \lambda_s$ may have repeated irreducible factors. The purpose of this section is to show that, over a

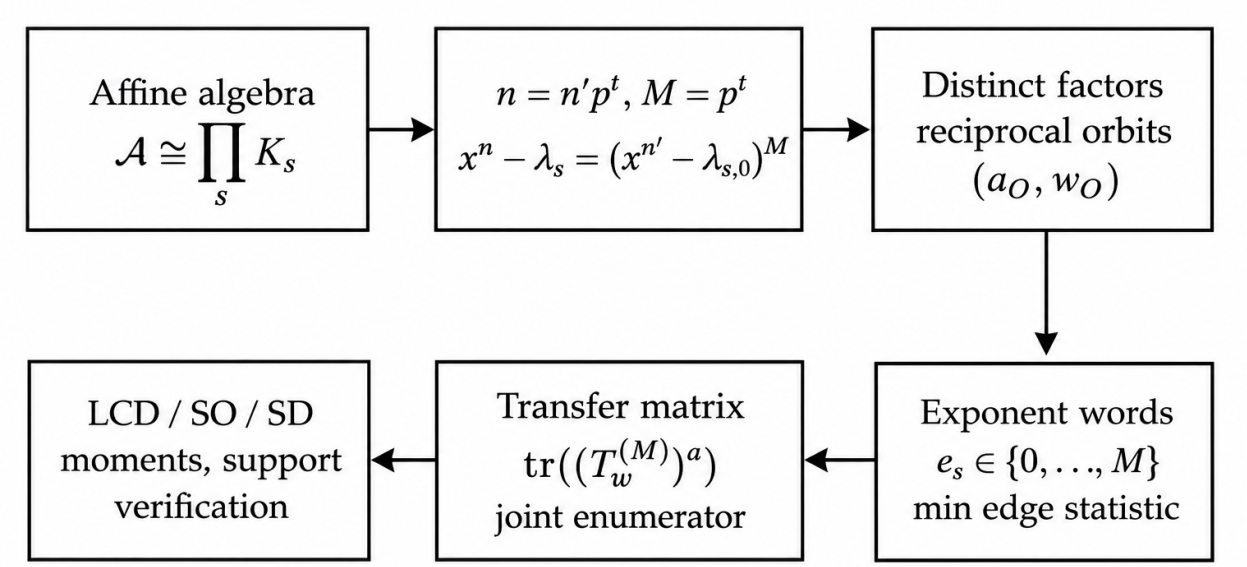


Fig. 1. Workflow of the paper.

finite field, every repeated-root modulus is the M-th power of a square-free kernel, where $M = p^{v_p(n)}$. A code is therefore determined by one exponent in $\{0, \ldots, M\}$ at each distinct irreducible factor of that kernel.

A. Square-free affine algebras and their components

Throughout, p is a prime and

$$q = p^e, \qquad e \geq 1.$$

For each component index s we write

$$K_s = \mathbb{F}_{q^{m_s}} = \mathbb{F}_{p^{d_s}}, \qquad d_s = em_s, \qquad m_s \geq 1.$$

The ambient algebra is

$$\mathcal{A} = \mathbb{F}_q[X_1, \ldots, X_\ell] \big/ \langle t_1(X_1), \ldots, t_\ell(X_\ell) \rangle, \tag{1}$$

where $\ell \geq 1$ and every t_i is monic, square-free, and of positive degree.

Proposition 1 (Square-free component decomposition)**.** *The algebra* (1) *has a fixed labeled decomposition*

$$\mathcal{A} \cong \prod_{s=1}^{N} K_s. \tag{2}$$

Proof. Each t_i is separable, so factoring it into pairwise coprime monic irreducibles and applying the Chinese remainder theorem exhibits $\mathbb{F}_q[X_i]/\langle t_i\rangle$ as a finite product of finite field extensions of $\mathbb{F}_q$. The algebra (1) is the tensor product over $\mathbb{F}_q$ of these one-variable quotients; a tensor product of finite étale algebras is finite étale, and a finite étale algebra over a finite field is a finite product of finite fields. Fixing the tensor factors and the Chinese remainder idempotents fixes the component labels in (2). □

We fix the decomposition (2) once and for all and write elements of $\mathcal{A}$ as tuples $(a_s)_s$. Let $\varepsilon_s \in \mathcal{A}$ be the primitive idempotent with 1 in component s and 0 elsewhere. Dimensions of global objects are taken over $\mathbb{F}_q$; a K_s-dimension is converted to an $\mathbb{F}_q$-dimension by multiplication by $m_s = [K_s : \mathbb{F}_q]$. Thus $\dim_{\mathbb{F}_q} \mathcal{A} = \sum_{s=1}^{N} m_s$.

It is important to keep the two uses of "square-free" separate: the algebra $\mathcal{A}$ is square-free (reduced), whereas from §II-C onwards the code modulus $x^n - \lambda_s$ is allowed, and intended, to be non-square-free.

B. Constacyclic codes over $\mathcal{A}$

Fix an arbitrary length $n \geq 1$ and a twist $\lambda = (\lambda_s)_{s=1}^{N} \in \mathcal{A}^*$, so $\lambda_s \in K_s^*$ for every s (throughout, $\mathcal{A}^*$ and K_s^* denote the unit groups). The λ-constacyclic shift on $\mathcal{A}^n$ is

$$T_\lambda(c_0, c_1, \ldots, c_{n-1}) = (\lambda c_{n-1}, c_0, \ldots, c_{n-2}).$$

Definition 2. An *$\mathcal{A}$-linear λ-constacyclic code* of length n over $\mathcal{A}$ is an $\mathcal{A}$-submodule $C \subseteq \mathcal{A}^n$ such that $T_\lambda(C) \subseteq C$.

Identify $(c_0, \ldots, c_{n-1})$ with $\sum_{i=0}^{n-1} c_i x^i$. Under this identification $\mathcal{A}^n$ becomes the quotient ring

$$R_{\mathcal{A},\lambda} = \mathcal{A}[x]/\langle x^n - \lambda\rangle,$$

multiplication by x corresponds to T_λ, and the $\mathcal{A}$-linear λ-constacyclic codes are exactly the ideals of $R_{\mathcal{A},\lambda}$. Coefficientwise, the decomposition (2) gives

$$R_{\mathcal{A},\lambda} \cong \prod_{s=1}^{N} K_s[x]/\langle x^n - \lambda_s\rangle.$$

Lemma 3. *Every $\mathcal{A}$-linear λ-constacyclic code decomposes uniquely as a product $C = \prod_{s=1}^{N} C_s$, where C_s is an ideal of $K_s[x]/\langle x^n - \lambda_s\rangle$. Conversely, every such product is an $\mathcal{A}$-linear λ-constacyclic code.*

Proof. Multiplication by ε_s projects $R_{\mathcal{A},\lambda}$ onto its s-th component ring. If C is an ideal, then $\varepsilon_s C$ is an ideal of the s-th component; since $1 = \sum_s \varepsilon_s$ is an orthogonal idempotent decomposition, $C = \bigoplus_s \varepsilon_s C$, which is the asserted product decomposition. The converse is immediate from the product ring structure. □

Remark 4. Definition 2 requires $\mathcal{A}$-linearity, not merely $\mathbb{F}_q$-linearity plus shift-invariance. This convention is essential for Lemma 3: subspaces that mix the labeled components are not counted in this paper.

Nothing in Definition 2 or Lemma 3 requires $\gcd(n, p) = 1$. The only place where the characteristic interacts with the length is the factorization of the component moduli, to which we now turn.

C. The repeated-root factorization

Write the length as

$$n = n' p^t, \qquad t = v_p(n) \geq 0, \qquad \gcd(n', p) = 1,$$

and put

$$M = p^t.$$

We call M the *multiplicity*; the reason is Lemma 5. For each s, the map $a \mapsto a^M = a^{p^t}$ is a power of Frobenius, hence an automorphism of K_s. Let $\lambda_{s,0}$ be the unique element of K_s^* such that

$$\lambda_{s,0}^M = \lambda_s, \qquad \text{i.e.} \qquad \lambda_{s,0} = \lambda_s^{p^{d_s - t \bmod d_s}} = \mathrm{Frob}^{-t}(\lambda_s). \tag{3}$$

Define the component modulus and its square-free kernel by

$$P_s(x) = x^n - \lambda_s, \qquad P_{s,0}(x) = x^{n'} - \lambda_{s,0}.$$

We deliberately write P_s for the modulus, since the letter M is reserved for the multiplicity.

Lemma 5 (Repeated-root factorization)**.** *For every component s,*

$$x^n - \lambda_s = \left(x^{n'} - \lambda_{s,0}\right)^M. \tag{4}$$

Moreover $x^{n'} - \lambda_{s,0}$ is square-free. Hence every distinct monic irreducible factor of $x^n - \lambda_s$ over K_s occurs with multiplicity exactly M.

Proof. In characteristic p, the Frobenius identity $(u-v)^{p^t} = u^{p^t} - v^{p^t}$ holds in any commutative K_s-algebra. Applying it to $u = x^{n'}$ and $v = \lambda_{s,0}$ and using (3) gives

$$\left(x^{n'} - \lambda_{s,0}\right)^M = x^{n'p^t} - \lambda_{s,0}^{p^t} = x^n - \lambda_s.$$

For square-freeness, the derivative of $P_{s,0}$ is $n'x^{n'-1}$ (equal to 1 when $n' = 1$). A common root α of $P_{s,0}$ and its derivative in an algebraic closure would satisfy $\alpha^{n'} = \lambda_{s,0} \neq 0$, hence $\alpha \neq 0$, and $n'\alpha^{n'-1} = 0$, contradicting $\gcd(n', p) = 1$. Thus $\gcd(P_{s,0}, P'_{s,0}) = 1$, so $P_{s,0}$ is separable and therefore square-free. Raising a square-free factorization to the M-th power gives every distinct irreducible factor multiplicity exactly M. □

Let

$$\mathcal{F}_s = \{\text{distinct monic irreducible factors of } P_{s,0} \text{ over } K_s\}.$$

By Lemma 5, $\mathcal{F}_s$ is also the set of distinct monic irreducible factors of P_s, and

$$P_s(x) = \prod_{f \in \mathcal{F}_s} f(x)^M.$$

The factorization theory needed here is exhausted by Lemma 5: over a finite field there are no other multiplicity patterns for a constacyclic modulus, and no hypothesis on the multiplicative order of λ_s is used.

Remark 6. The generator of a code divides $x^n - \lambda_s$, but every computation (factor set, reciprocal action, examples) is performed with $x^{n'} - \lambda_{s,0}$. Accordingly, every example in this paper records both $\lambda_{s,0}$ and $\lambda_s = \lambda_{s,0}^M$.

D. Exponent selections

Fix a component s. An *exponent selection* is a tuple $\mathbf{e} = (e_f)_{f \in \mathcal{F}_s}$ with integers

$$0 \leq e_f \leq M \qquad (f \in \mathcal{F}_s).$$

(The bold $\mathbf{e}$ denotes the selection; its coordinates are the e_f. The plain letter e continues to denote the exponent in $q = p^e$.) Set

$$g_{\mathbf{e}} = \prod_{f \in \mathcal{F}_s} f^{e_f}, \qquad h_{\mathbf{e}} = \frac{P_s}{g_{\mathbf{e}}} = \prod_{f \in \mathcal{F}_s} f^{M - e_f},$$

and let $C_s(\mathbf{e}) = \langle g_{\mathbf{e}} \rangle \subseteq K_s[x]/\langle P_s \rangle$.

Lemma 7. *The λ_s-constacyclic codes in $K_s[x]/\langle P_s\rangle$ are exactly the ideals $C_s(\mathbf{e})$, and distinct exponent selections give distinct codes. Moreover*

$$\dim_{K_s} C_s(\mathbf{e}) = n - \sum_{f\in\mathcal{F}_s} e_f \deg f = \sum_{f\in\mathcal{F}_s} (M - e_f)\deg f. \tag{5}$$

Proof. Since $K_s[x]$ is a principal ideal domain, the ideals of the quotient $K_s[x]/\langle P_s\rangle$ are the principal ideals generated by the monic divisors of P_s, each monic divisor determining a distinct ideal. By Lemma 5, a monic divisor of $P_s = \prod_{f\in\mathcal{F}_s} f^M$ is precisely a choice of exponents $e_f \in \{0,\dots,M\}$, and unique factorization makes the correspondence bijective. The ideal generated by a polynomial of degree d in a quotient by a degree-n modulus has K_s-dimension $n-d$; with $d = \deg g_{\mathbf{e}} = \sum_f e_f \deg f$ and $n = M\sum_f \deg f$ this gives both forms in (5). □

Globally, write a selection as $\mathbf{e} = (\mathbf{e}^{(s)})_{s=1}^N$ and put $C(\mathbf{e}) = \prod_{s=1}^N C_s(\mathbf{e}^{(s)})$. Set

$$N_F = \sum_{s=1}^N |\mathcal{F}_s|, \qquad W = n\sum_{s=1}^N m_s = \dim_{\mathbb{F}_q} \mathcal{A}^n.$$

For a global code C we write $\kappa(C) = \dim_{\mathbb{F}_q} C$.

Proposition 8. *The map $\mathbf{e} \mapsto C(\mathbf{e})$ is a bijection from $\prod_{s=1}^N \{0,\dots,M\}^{\mathcal{F}_s}$ to the set of $\mathcal{A}$-linear λ-constacyclic codes of length n over $\mathcal{A}$. In particular the number of labeled codes is $(M+1)^{N_F}$, and*

$$\kappa(C(\mathbf{e})) = \sum_{s=1}^N \sum_{f\in\mathcal{F}_s} m_s(\deg f)\,(M - e_f). \tag{6}$$

Proof. By Lemma 3 a global code is a product of component ideals, and by Lemma 7 each component ideal is determined by a unique exponent selection. This proves the bijection and the count $(M+1)^{N_F}$. Formula (6) is (5) summed over components, with each K_s-dimension multiplied by m_s. □

The extreme selections are $\mathbf{e} = (M,\dots,M)$, giving the zero code, and $\mathbf{e} = (0,\dots,0)$, giving the whole ambient space, of $\mathbb{F}_q$-dimension W. Counting throughout is *labeled*: distinct exponent selections are counted separately, with no quotient by any equivalence relation. In probabilistic statements, "uniform" always refers to the uniform measure on the $(M+1)^{N_F}$ exponent selections.

Thus each distinct irreducible factor contributes one exponent coordinate, with the $M+1$ possible values $0, 1, \dots, M$.

III. k-Galois duality

This section develops k-Galois duality directly for the exponent selections of Proposition 8. The Euclidean dual generator is verified for arbitrary length in Lemma 14. The outcome is an explicit compatibility condition for each component; under compatibility, the reciprocal permutes the distinct kernel factors, and multiplicities enter the dual generator through the identity $(f^m)^{\#} = (f^{\#})^m$.

A. Pairings, duals and hulls

Fix an integer k with $0 \le k < e$. On K_s put

$$\sigma_{s,k}(a) = a^{p^k}, \qquad \rho_{s,k}(a) = a^{p^{d_s-k}} = \sigma_{s,k}^{-1}(a).$$

The *component k-Galois pairing* on K_s^n is

$$\langle x, y\rangle_{s,k} = \sum_{i=0}^{n-1} x_i\, \sigma_{s,k}(y_i) = \sum_{i=0}^{n-1} x_i y_i^{p^k}. \tag{7}$$

It is K_s-linear in the first argument and $\sigma_{s,k}$-semilinear in the second; note that it is *code-first* in the sense of Remark 22. For a K_s-linear code $C \subseteq K_s^n$ define

$$D_k(C) = \{y \in K_s^n : \langle c, y\rangle_{s,k} = 0 \text{ for every } c \in C\},$$

$$\mathrm{Hull}_k(C) = C \cap D_k(C).$$

the *k-Galois dual* and *k-Galois hull*. The annihilator $D_k(C)$ is a K_s-subspace: if $y \in D_k(C)$ and $a \in K_s$, then $\langle c, ay\rangle_{s,k} = \sigma_{s,k}(a)\langle c, y\rangle_{s,k} = 0$.

On the s-th component put $\sigma_{\mathcal{A},k}|_{K_s} = \sigma_{s,k}$ and $\rho_{\mathcal{A},k}|_{K_s} = \rho_{s,k}$; these componentwise maps are mutually inverse ring automorphisms of $\mathcal{A}$. The *global k-Galois pairing* on $\mathcal{A}^n$ is the $\mathcal{A}$-valued pairing

$$\langle x, y\rangle_{\mathcal{A},k} = \sum_{i=0}^{n-1} x_i\, \sigma_{\mathcal{A},k}(y_i) = \Big(\sum_{i=0}^{n-1} x_{s,i} y_{s,i}^{p^k}\Big)_{s=1}^{N}.$$

The global dual and hull $D_{\mathcal{A},k}(C)$ and $\mathrm{Hull}_{\mathcal{A},k}(C)$ are defined by the same annihilator formulas with $\langle\cdot,\cdot\rangle_{\mathcal{A},k}$. For a global code C we write

$$\eta_k(C) = \dim_{\mathbb{F}_q} \mathrm{Hull}_{\mathcal{A},k}(C).$$

Lemma 9. *If $C = \prod_{s=1}^{N} C_s$ is a product code, then*

$$D_{\mathcal{A},k}(C) = \prod_{s=1}^{N} D_k(C_s), \qquad \mathrm{Hull}_{\mathcal{A},k}(C) = \prod_{s=1}^{N} \mathrm{Hull}_k(C_s),$$

and consequently $\eta_k(C) = \sum_{s=1}^{N} m_s \dim_{K_s} \mathrm{Hull}_k(C_s)$.

Proof. An element of $\mathcal{A}$ is zero if and only if every component is zero, so $\langle c, y\rangle_{\mathcal{A},k} = 0$ if and only if $\langle c_s, y_s\rangle_{s,k} = 0$ for all s. Taking for c a codeword supported in a single component shows $y_s \in D_k(C_s)$ for every s; the converse inclusion is immediate. Intersections of product sets are products of intersections, giving the hull statement, and the dimension formula is the usual conversion by m_s. □

Lemma 9 is a product argument and is indifferent to whether p divides n; the same is true of the next statement, which reduces k-Galois duality to Euclidean duality.

Proposition 10. *For every K_s-linear code $C \subseteq K_s^n$,*

$$D_k(C) = \rho_{s,k}\big(C^{\perp_E}\big),$$

where $C^{\perp_E}$ is the ordinary Euclidean dual and $\rho_{s,k}$ acts coordinatewise.

Proof. The k-Galois dual $D_k(C)$ is the set of vectors y satisfying $\sum_i c_i \sigma_{s,k}(y_i) = 0$ for all $c \in C$; with $z = \sigma_{s,k}(y)$ this says $z \in C^{\perp_E}$, and applying $\rho_{s,k}$ coordinatewise gives $y = \rho_{s,k}(z)$. □

Remark 11. For $k = 0$ the pairing (7) is Euclidean on every component. On $K_s = \mathbb{F}_{p^{d_s}}$ it is Hermitian if and only if d_s is even and $k \equiv d_s/2 \pmod{d_s}$; in particular, when e is even the base-field choice $k = e/2$ is Hermitian only on components with $m_s = 1$, not in general. We therefore avoid the unqualified phrase "the Hermitian case".

We use the standard terminology, componentwise or globally according to context: a code is *k-Galois LCD* (linear with complementary dual) if $\mathrm{Hull}_k(C) = \{0\}$, *self-orthogonal* if $C \subseteq D_k(C)$, *dual-containing* if $D_k(C) \subseteq C$, and *self-dual* if $C = D_k(C)$ as subspaces (no involutivity of D_k is assumed).

B. The inverse-Frobenius reciprocal

Let $f(x) = \sum_{i=0}^{r} f_i x^i \in K_s[x]$ be monic with $f_0 \neq 0$. The *normalized inverse-Frobenius reciprocal* of f is

$$f^{\#_{s,k}}(x) = \rho_{s,k}(f_0)^{-1} \sum_{i=0}^{r} \rho_{s,k}(f_i) x^{r-i}. \tag{8}$$

For $k = 0$ this is the ordinary normalized reciprocal, which we also denote by f^* when the field is arbitrary.

Lemma 12. *The map $f \mapsto f^{\#_{s,k}}$ on monic polynomials with nonzero constant term is monic and degree-preserving, is multiplicative, hence satisfies $(f^m)^{\#_{s,k}} = (f^{\#_{s,k}})^m$ for every $m \geq 0$, preserves irreducibility, and is inverted by the corresponding normalized reciprocal with $\sigma_{s,k}$ in place of $\rho_{s,k}$.*

Proof. Reversal puts $\rho_{s,k}(f_0) \neq 0$ in degree r, so the normalization in (8) makes $f^{\#_{s,k}}$ monic of degree r; its constant term is $\rho_{s,k}(f_r)/\rho_{s,k}(f_0) = \rho_{s,k}(f_0)^{-1} \neq 0$. If $\widetilde{f} = \sum_i \rho_{s,k}(f_i)x^{r-i}$ denotes the unnormalized reversal, then $\widetilde{fg} = \widetilde{f}\,\widetilde{g}$ for monic f, g with nonzero constant terms, because coefficient reversal and the ring homomorphism $\rho_{s,k}$ are both multiplicative; the normalizing constants multiply as $\rho_{s,k}(f_0 g_0)^{-1} = \rho_{s,k}(f_0)^{-1}\rho_{s,k}(g_0)^{-1}$, proving multiplicativity and in particular the power identity. Applying instead the normalized reciprocal built from $\sigma_{s,k}$ reverses the coefficients a second time and applies $\sigma_{s,k}\rho_{s,k} = \mathrm{id}$; since the constant term of $f^{\#_{s,k}}$ is $\rho_{s,k}(f_0)^{-1}$, the second normalization restores f, so the map is bijective with the stated inverse. Finally, a nontrivial monic factorization $f = gh$ maps under the multiplicative bijection to a nontrivial factorization of $f^{\#_{s,k}}$, and conversely via the inverse; hence irreducibility is preserved. □

Lemma 13. *Let f be as above and let α be a root of f in an extension field. Then $\alpha^{-p^{d_s-k}}$ is a root of $f^{\#_{s,k}}$.*

Proof. Since $f_0 \neq 0$ we have $\alpha \neq 0$. Extend $\rho = \rho_{s,k}$ to the extension field by $\rho(a) = a^{p^{d_s-k}}$. From (8),

$$f^{\#_{s,k}}(x) = \rho(f_0)^{-1}x^r \sum_{i=0}^{r} \rho(f_i)x^{-i}.$$

Put $\beta = \rho(\alpha^{-1}) = \alpha^{-p^{d_s-k}}$, so $\beta^{-1} = \rho(\alpha)$. Then

$$\sum_{i=0}^{r} \rho(f_i)\beta^{-i} = \sum_{i=0}^{r} \rho(f_i)\rho(\alpha)^i = \rho\Big(\sum_{i=0}^{r} f_i\alpha^i\Big) = \rho(f(\alpha)) = 0,$$

hence $f^{\#_{s,k}}(\beta) = 0$. □

C. Euclidean and k-Galois dual generators

We first verify the Euclidean dual generator for arbitrary length n.

Lemma 14. *Let K be a finite field, $a \in K^*$ and $n \geq 1$ arbitrary. Let $G \mid x^n - a$ be monic, put $H = (x^n - a)/G$, and let H^* be the ordinary normalized reciprocal of H. Then the Euclidean dual of the ideal $\langle G\rangle$ in $K[x]/\langle x^n - a\rangle$ is the a^{-1}-constacyclic ideal generated by H^*.*

Proof. Write $r = \deg G$ and $d_H = \deg H$, so $r + d_H = n$; note $H(0) \neq 0$ since $G(0)H(0) = -a \neq 0$, so H^* is defined (and equals 1 if $d_H = 0$). The non-wrapped shifts $x^i G$, $0 \leq i < d_H$, form a basis of $\langle G \rangle$, and we compare them with the non-wrapped shifts $x^j H^*$, $0 \leq j < r$. The Euclidean pairing of $x^i G$ and $x^j H^*$ is, up to the nonzero scalar $H(0)^{-1}$, the coefficient of $x^{d_H - i + j}$ in GH: indeed, reversal sends the coefficient H_u of H to degree $d_H - u$ in H^*, so aligned terms in the pairing have $u + v = d_H - i + j$ where v indexes coefficients of G. Since $1 \leq d_H - i + j \leq n - 1$ and $GH = x^n - a$ has no terms in these degrees, the pairing is zero. Thus the two shift spaces are mutually orthogonal.

By multiplicativity of the ordinary reciprocal and $(x^n - a)^* = x^n - a^{-1}$ we have $H^* \mid x^n - a^{-1}$, so the shifts $x^j H^*$ span the a^{-1}-constacyclic ideal generated by H^*, of dimension r. This equals the codimension of $\langle G \rangle$; an orthogonal subspace of that dimension is the dual. The boundary cases $r = 0$ and $d_H = 0$ are included: they give the zero ideal generated by $H^* = x^n - a^{-1}$ in its quotient, and the whole space generated by $H^* = 1$, respectively.

No step uses that $x^n - a$ is square-free, or that $\gcd(n, \operatorname{char} K) = 1$: the argument sees only the identity $GH = x^n - a$ and a dimension count, and any multiplicities are carried along inside H. □

Applying Proposition 10 to Lemma 14 gives the k-Galois dual generator. Note that applying $\rho_{s,k}$ to the coefficients of the ordinary reciprocal h^* gives exactly $h^{\#_{s,k}}$ as defined in (8).

Theorem 15. *Let $C_s(\mathbf{e}) = \langle g_{\mathbf{e}} \rangle$ be a component code and $h_{\mathbf{e}} = P_s / g_{\mathbf{e}} = \prod_{f \in \mathcal{F}_s} f^{M - e_f}$. Then $D_k(C_s(\mathbf{e}))$ is the constacyclic ideal generated by $h_{\mathbf{e}}^{\#_{s,k}}$, with dual twist*

$$\mu_s = \lambda_s^{-p^{d_s - k}}.$$

Proof. By Proposition 10, $D_k(C_s(\mathbf{e})) = \rho_{s,k}(C_s(\mathbf{e})^{\perp_E})$ with $\rho_{s,k}$ acting coordinatewise, i.e. coefficientwise on polynomial representatives. By Lemma 14 the Euclidean dual is the λ_s^{-1}-constacyclic ideal generated by $h_{\mathbf{e}}^*$. Coefficientwise application of $\rho_{s,k}$ sends generator $h_{\mathbf{e}}^*$ to $h_{\mathbf{e}}^{\#_{s,k}}$ and twist λ_s^{-1} to $\rho_{s,k}(\lambda_s^{-1}) = \lambda_s^{-p^{d_s - k}} = \mu_s$. □

D. Compatibility and incompatible twists

The dual twist in Theorem 15 need not equal λ_s. We now determine exactly when it does. Since all factor computations take place in the base modulus $P_{s,0}$, we first compute its reciprocal: directly from (8),

$$P_{s,0}^{\#_{s,k}} = x^{n'} - \lambda_{s,0}^{-p^{d_s - k}}. \tag{9}$$

Call the component s *compatible* (for the given k) if $f \mapsto f^{\#_{s,k}}$ permutes $\mathcal{F}_s$, and call λ *compatible* if every component is.

Proposition 16. *For each component s the following are equivalent:*

(i) *s is compatible;*
(ii) $\lambda_{s,0}^{1+p^{d_s-k}} = 1$*;*
(iii) $\lambda_s^{1+p^{d_s-k}} = 1$.

In particular the compatibility condition is the same for $\lambda_{s,0}$ and for $\lambda_s = \lambda_{s,0}^M$.

Proof. Let $f \in \mathcal{F}_s$. By Lemma 12, $f^{\#_{s,k}}$ is monic irreducible, and since $f \mid P_{s,0}$, multiplicativity gives $f^{\#_{s,k}} \mid P_{s,0}^{\#_{s,k}}$, which by (9) is $x^{n'} - \lambda_{s,0}^{-p^{d_s-k}}$. If (ii) holds, then $\lambda_{s,0}^{-p^{d_s-k}} = \lambda_{s,0}$, so $P_{s,0}^{\#_{s,k}} = P_{s,0}$ and $f^{\#_{s,k}} \in \mathcal{F}_s$; the inverse map of Lemma 12 then makes $f \mapsto f^{\#_{s,k}}$ a permutation of $\mathcal{F}_s$, giving (i). Conversely, if (i) holds, applying $\#_{s,k}$ to the square-free product $P_{s,0} = \prod_{f\in\mathcal{F}_s} f$ gives $P_{s,0}^{\#_{s,k}} = \prod_{f\in\mathcal{F}_s} f^{\#_{s,k}} = P_{s,0}$; comparing with (9) yields $\lambda_{s,0}^{-p^{d_s-k}} = \lambda_{s,0}$, which is (ii).

For (ii)$\Leftrightarrow$(iii), put $c = 1 + p^{d_s-k}$. Then $\lambda_s^c = (\lambda_{s,0}^M)^c = (\lambda_{s,0}^c)^M$. The map $y \mapsto y^M = y^{p^t}$ on K_s^* is a Frobenius power, hence injective with $1^M = 1$; therefore $(\lambda_{s,0}^c)^M = 1$ if and only if $\lambda_{s,0}^c = 1$. □

Here $d_s = em_s$ is the absolute degree of K_s over $\mathbb{F}_p$; when $m_s = 1$ this reduces to the base-field condition $\lambda_{s,0}^{1+p^{e-k}} = 1$.

Under compatibility we define

$$\tau_{s,k} : \mathcal{F}_s \longrightarrow \mathcal{F}_s, \qquad \tau_{s,k}(f) = f^{\#_{s,k}},$$

a degree-preserving permutation by Lemma 12; it need not be an involution. Condition (iii) is exactly $\mu_s = \lambda_s$, so by Theorem 15 the k-Galois dual is again λ_s-constacyclic, and its generator can be read off factor by factor.

Proposition 17. *If s is compatible, then for every exponent selection $\mathbf{e}$,*

$$h_{\mathbf{e}}^{\#_{s,k}} = \prod_{f\in\mathcal{F}_s} f^{\,M-e_{\tau_{s,k}^{-1}(f)}}. \tag{10}$$

In words: the exponent of f in the dual generator is M minus the exponent of its $\tau_{s,k}$-predecessor in $g_{\mathbf{e}}$.

Proof. By multiplicativity and the power identity in Lemma 12,

$$h_{\mathbf{e}}^{\#_{s,k}} = \prod_{f\in\mathcal{F}_s} \left(f^{M-e_f}\right)^{\#_{s,k}} = \prod_{f\in\mathcal{F}_s} \tau_{s,k}(f)^{\,M-e_f}.$$

Re-indexing by $g = \tau_{s,k}(f)$ gives (10). □

Remark 18. The twists $\lambda_s = 1$ (cyclic) and $\lambda_s = -1$ (negacyclic) are always compatible: for odd p, $(-1)^{1+p^{d_s-k}} = 1$ because p^{d_s-k} is odd, and for $p = 2$ one has $-1 = 1$.

It remains to dispose of the incompatible components. They turn out to be enumeration-theoretically trivial.

Proposition 19. *If the component s is incompatible, then* $\mathrm{Hull}_k(C_s(\mathbf{e})) = \{0\}$ *for every exponent selection* $\mathbf{e}$*; in particular every such component code is k-Galois LCD.*

Proof. By Proposition 16 and Theorem 15, incompatibility means $\mu_s = \lambda_s^{-p^{d_s-k}} \neq \lambda_s$, the generator $g_{\mathbf{e}}$ divides $x^n - \lambda_s$, and the dual generator $h_{\mathbf{e}}^{\#_{s,k}}$ divides $x^n - \mu_s$. The two moduli differ by the nonzero constant $\mu_s - \lambda_s$, hence are coprime; therefore $\gcd(g_{\mathbf{e}}, h_{\mathbf{e}}^{\#_{s,k}}) = 1$. Moreover $\deg g_{\mathbf{e}} + \deg h_{\mathbf{e}}^{\#_{s,k}} = \deg g_{\mathbf{e}} + \deg h_{\mathbf{e}} = n$. A hull codeword has a unique representative polynomial of degree $< n$; that representative is divisible in $K_s[x]$ both by $g_{\mathbf{e}}$ (since it lies in $C_s(\mathbf{e})$) and by $h_{\mathbf{e}}^{\#_{s,k}}$ (since it lies in the dual, whose words are the multiples of its generator of degree $< n$). Being divisible by two coprime polynomials whose degrees sum to n, it is divisible by a polynomial of degree n, hence is zero. □

Corollary 20. *Let s be incompatible. Then the component contributes* 0 *to every global hull dimension. The only self-orthogonal component code is* $\{0\}$*, the only dual-containing component code is K_s^n, and there is no self-dual component code.*

Proof. The first statement follows from Proposition 19 and Lemma 9. When $C_s(\mathbf{e})$ is self-orthogonal we have $C_s(\mathbf{e}) = \mathrm{Hull}_k(C_s(\mathbf{e})) = \{0\}$. When it is dual-containing, $D_k(C_s(\mathbf{e})) = \mathrm{Hull}_k(C_s(\mathbf{e})) = \{0\}$, and by Proposition 10 $\dim_{K_s} D_k(C_s(\mathbf{e})) = n - \dim_{K_s} C_s(\mathbf{e})$, forcing $C_s(\mathbf{e}) = K_s^n$. Self-duality would give $C_s(\mathbf{e}) = \mathrm{Hull}_k(C_s(\mathbf{e})) = \{0\}$ and at the same time $D_k(C_s(\mathbf{e})) = \{0\}$, impossible for $n \geq 1$. □

Example 21. Let $K_s = \mathbb{F}_5$, $n' = 2$, $M = 5$, so $n = 10$, and take $k = 0$ and $\lambda_{s,0} = \lambda_s = 2$. Then $x^2 - 2$ is irreducible over $\mathbb{F}_5$ (the nonzero squares are $1, 4$), so $\mathcal{F}_s = \{x^2 - 2\}$, and $\lambda_s^{1+5} = 2^6 = 4 \neq 1$ in $\mathbb{F}_5$: the component is incompatible. The six codes $\langle (x^2-2)^e \rangle$, $e = 0, \ldots, 5$, therefore have $(\kappa, \eta_0) = (10 - 2e, 0)$ by Proposition 19.

Remark 22. Part of the literature uses the *candidate-first* dual $D_{\mathrm{cand}}(C) = \{y : \sum_i y_i \sigma_{s,k}(c_i) = 0 \text{ for all } c \in C\}$. Substituting $y = \sigma_{s,k}(z)$ gives $\sum_i y_i \sigma_{s,k}(c_i) = \sigma_{s,k}(\sum_i z_i c_i)$, hence $D_{\mathrm{cand}}(C) =$

$\sigma_{s,k}(C^{\perp_E})$, and therefore $D_{\mathrm{cand}}(C) = \sigma_{s,k}^2(D_k(C))$ by Proposition 10. The two duals are different subspaces in general; generator and factor-support statements in this paper refer to the code-first dual (7). Hull *dimensions*, however, agree: if B is a row basis of C and $G = B\,\sigma_{s,k}(B)^{\mathsf{T}}$, then for $y = vB$ the code-first hull condition is $G\,\sigma_{s,k}(v)^{\mathsf{T}} = 0$ and the candidate-first condition is $vG = 0$; right and left nullspaces have the same dimension $\dim C - \operatorname{rank} G$. Thus every hull-dimension statement below is convention-independent, and the Gram-matrix rank identity just displayed is what our independent computational verification in Section VI implements.

We have now covered every twist λ: incompatible components contribute no hull dimension at all, and compatible components are governed by the permutation $\tau_{s,k}$ and the dual exponents (10). The next section turns this into an explicit formula for $\eta_k(C(\mathbf{e}))$ and then into the joint enumerator.

IV. Hull statistic and the joint enumerator

We now convert the dual exponents of Section III into a formula for the hull dimension and then into the joint enumerator. Compatible components are organized by the cycles of $\tau_{s,k}$; incompatible components require no orbit structure, because their hull contribution is identically zero (Proposition 19).

A. Reciprocal orbits and exponent words

Let s be a compatible component. Denote by $\mathcal{O}_s$ the set of cycles (orbits) of the permutation $\tau_{s,k}$ on the *distinct* factor set $\mathcal{F}_s$; multiplicities do not create new factors. For $O \in \mathcal{O}_s$ write

$$a_O = |O|, \quad d_O := \deg(g), \text{for all } g \in O, \quad w_O = m_s d_O.$$

The degree is constant on O by Lemma 12. Enumerate

$$O = (f_{O,1}, \ldots, f_{O,a_O}) \qquad \text{so that} \qquad \tau_{s,k}(f_{O,i}) = f_{O,i+1},$$

with indices modulo a_O. The starting point of this enumeration is arbitrary; its cyclic direction is the one prescribed by $\tau_{s,k}$.

The restriction of an exponent selection to O is the *exponent word*

$$(e_{O,1}, \ldots, e_{O,a_O}) \in \{0, \ldots, M\}^{a_O}, \qquad e_{O,i} = e_{f_{O,i}}.$$

By Proposition 8, a global selection is an independent choice of one exponent word on every compatible orbit, together with one exponent at every distinct factor of every incompatible component. This independence is the engine of the factorization in Theorem 28.

For an incompatible component and $f \in \mathcal{F}_s$ we use the individual weight $w_f = m_s \deg f$; on a compatible orbit, $w_f = w_O$ for every $f \in O$.

B. The hull generator and the min*-edge formula*

Lemma 23. *Let s be compatible and let $\mathbf{e}$ be an exponent selection. Then* $\mathrm{Hull}_k(C_s(\mathbf{e}))$ *is the ideal generated by* $\mathrm{lcm}(g_{\mathbf{e}}, h_{\mathbf{e}}^{\#_{s,k}})$*. On an orbit $O = (f_{O,1}, \dots, f_{O,a_O})$ the exponents are:*

polynomial	*exponent at $f_{O,i}$*
$g_{\mathbf{e}}$	$e_{O,i}$
$h_{\mathbf{e}}$	$M - e_{O,i}$
$h_{\mathbf{e}}^{\#_{s,k}}$	$M - e_{O,i-1}$
$\mathrm{lcm}(g_{\mathbf{e}}, h_{\mathbf{e}}^{\#_{s,k}})$	$\max\{e_{O,i},\, M - e_{O,i-1}\}$

with indices modulo a_O.

Proof. Under compatibility, Theorem 15 says that both $C_s(\mathbf{e})$ and $D_k(C_s(\mathbf{e}))$ are ideals in the same quotient $K_s[x]/\langle P_s\rangle$, generated by the monic divisors $g_{\mathbf{e}}$ and $h_{\mathbf{e}}^{\#_{s,k}}$ of P_s. A polynomial representative of degree $< n$ lies in both ideals if and only if it is divisible in $K_s[x]$ by both generators, equivalently by their least common multiple. That least common multiple again divides P_s, since the exponent of each distinct factor is a maximum of two numbers in $\{0, \dots, M\}$. The table combines the definitions of $g_{\mathbf{e}}$ and $h_{\mathbf{e}}$ with Proposition 17, using $\tau_{s,k}^{-1}(f_{O,i}) = f_{O,i-1}$. □

Proposition 24. *For every exponent selection $\mathbf{e}$,*

$$\eta_k(C(\mathbf{e})) = \sum_{s \text{ compatible}} \sum_{O \in \mathcal{O}_s} w_O \sum_{i=1}^{a_O} \min\{e_{O,i},\, M - e_{O,i+1}\}, \tag{11}$$

whereas

$$\begin{aligned} \kappa(C(\mathbf{e})) = &\sum_{s \text{ compatible}} \sum_{O \in \mathcal{O}_s} w_O \sum_{i=1}^{a_O} (M - e_{O,i}) \\ &+ \sum_{s \text{ incompatible}} \sum_{f \in \mathcal{F}_s} w_f (M - e_f). \end{aligned} \tag{12}$$

In particular, if every component is compatible, the second sum in (12) *is absent.*

Proof. Let s be compatible. By Lemma 23 and the dimension formula for a principal ideal, the contribution of the factor $f_{O,i}$ to $\dim_{K_s} \mathrm{Hull}_k(C_s(\mathbf{e}))$ is

$$d_O\Big(M - \max\{e_{O,i}, M - e_{O,i-1}\}\Big) = d_O \min\{e_{O,i-1}, M - e_{O,i}\}.$$

Re-indexing $i-1 \mapsto i$ gives $d_O \min\{e_{O,i}, M - e_{O,i+1}\}$. Multiplying by m_s, summing over i and over all orbits of all compatible components, and using Proposition 19 for the remaining components gives (11) (via Lemma 9). Formula (12) is (6) regrouped by orbits on compatible components. □

Remark 25. For a single compatible orbit, (11) agrees with the one-code hull-dimension formula of Debnath et al. [23] on the range where both apply. Their expression is in the equivalent "max" form $aM - \sum_i \max\{e_i, M - e_{i-1}\}$, and the identity $M - \max\{u, v\} = \min\{M-u, M-v\}$ converts it, term by term, into (11) (the hull dimension is unaffected by the candidate-first/code-first choice of dual convention; see Remark 22). No coprimality condition $\gcd(\mathrm{ord}\,\lambda_s, n') = 1$ is needed here: only the component compatibility condition $\lambda_{s,0}^{1+p^{d_s-k}} = 1$ matters, and components failing it have trivial hull by Proposition 19. In [23] the compatibility and factorization hypotheses are stated over the base field $\mathbb{F}_q$ ($q = p^e$) as $\mathrm{ord}\,\lambda_s \mid 1 + p^{e-k}$ and $\gcd(\mathrm{ord}\,\lambda_s, n') = 1$. For a component field K_s of absolute degree $d_s = em_s$ over $\mathbb{F}_p$ our criterion uses d_s in place of e: the family $\mathbb{F}_{64}$ over $\mathbb{F}_8$ with $k = 1$, $n' = 3$, $M = 2$ and $\lambda_{s,0}$ of order 3 is compatible by our criterion ($3 \mid 1 + 2^5 = 33$) but not by the base-field criterion ($3 \nmid 1 + 2^2 = 5$), and $\gcd(\mathrm{ord}\,\lambda_{s,0}, n') = 3 \neq 1$. It is included in the independent brute-force checks of Section VI-C.

Thus, on one compatible orbit, κ is a sum over the *vertices* of a directed cycle and η_k is a sum over its *directed edges* $i \to i+1$, both scaled by w_O.

C. Transfer matrix and the main theorem

Definition 26. The *joint code-dimension and k-Galois-hull enumerator* is

$$\mathscr{E}^{(M)}(U, Z) := \sum_{\mathbf{e}} U^{\kappa(C(\mathbf{e}))} Z^{\eta_k(C(\mathbf{e}))} \in \mathbb{Z}[U, Z],$$

the sum running over all $(M+1)^{N_F}$ exponent selections. We write $N_{\kappa,\eta} = [U^\kappa Z^\eta]\mathscr{E}^{(M)}(U, Z)$.

By Proposition 8, $N_{\kappa,\eta}$ is the number of labeled $\mathcal{A}$-linear λ-constacyclic codes C with $\dim_{\mathbb{F}_q} C = \kappa$ and $\dim_{\mathbb{F}_q} \mathrm{Hull}_k(C) = \eta$.

Fix a compatible orbit of length a and weight w. Its one-orbit polynomial is

$$\mathscr{E}^{(M)}_{a,w}(U,Z) := \sum_{(e_1,\dots,e_a)\in\{0,\dots,M\}^a} U^{w\sum_{i=1}^{a}(M-e_i)} \times Z^{w\sum_{i=1}^{a}\min\{e_i,M-e_{i+1}\}}. \tag{13}$$

indices modulo a. Define the $(M+1)\times(M+1)$ transfer matrix, with rows and columns indexed by states $0,\dots,M$, by

$$\big(T^{(M)}_w(U,Z)\big)_{\alpha,\beta} = U^{w(M-\beta)}Z^{w\min\{\alpha,M-\beta\}}, \qquad 0\le\alpha,\beta\le M. \tag{14}$$

The entry records the destination vertex β in the U-exponent and the directed edge $\alpha\to\beta$ in the Z-exponent. For example,

$$T^{(2)}_w(U,Z) = \begin{pmatrix} U^{2w} & U^{w} & 1 \\ U^{2w}Z^{w} & U^{w}Z^{w} & 1 \\ U^{2w}Z^{2w} & U^{w}Z^{w} & 1 \end{pmatrix}.$$

Lemma 27. *For every $a\ge 1$,*

$$\operatorname{tr}\big(T^{(M)}_w(U,Z)^a\big) = \mathscr{E}^{(M)}_{a,w}(U,Z).$$

Proof. A diagonal entry of $T^{(M)}_w(U,Z)^a$ is a sum over closed walks $e_1\to e_2\to\cdots\to e_a\to e_1$ on the state set $\{0,\dots,M\}$. The product of the edge weights along the walk is

$$\prod_{i=1}^{a} U^{w(M-e_{i+1})}Z^{w\min\{e_i,M-e_{i+1}\}} = U^{w\sum_{i=1}^{a}(M-e_i)}Z^{w\sum_{i=1}^{a}\min\{e_i,M-e_{i+1}\}}.$$

because each vertex occurs exactly once as a destination. Summing over all diagonal entries is summing over all cyclic words $(e_1,\dots,e_a)$, which is (13). □

Let

$$\mathcal{S}^{\mathrm{comp}}_k = \{s : \lambda_s^{1+p^{d_s}-k} = 1\}, \qquad \mathcal{S}^{\mathrm{inc}}_k = \{1,\dots,N\}\setminus\mathcal{S}^{\mathrm{comp}}_k.$$

Theorem 28. *For every $n\ge 1$, every $\lambda\in\mathcal{A}^*$, and every $0\le k<e$,*

$$\mathscr{E}^{(M)}(U,Z) = \prod_{s\in\mathcal{S}^{\mathrm{comp}}_k}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\Big(T^{(M)}_{w_O}(U,Z)^{a_O}\Big) \times \prod_{s\in\mathcal{S}^{\mathrm{inc}}_k}\prod_{f\in\mathcal{F}_s}\Big(1+U^{w_f}+U^{2w_f}+\cdots+U^{Mw_f}\Big). \tag{15}$$

where an empty product is 1*. In particular, if every component is compatible, then*

$$\mathscr{E}^{\circ(M)}(U,Z)=\prod_{s=1}^{N}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\Big(T_{w_O}^{(M)}(U,Z)^{a_O}\Big).$$

In all cases, $[U^{\kappa}Z^{\eta}]\mathscr{E}^{\circ(M)}(U,Z)$ *is the exact number of labeled* λ*-constacyclic codes with* $\mathbb{F}_q$*-dimension* κ *and k-Galois-hull* $\mathbb{F}_q$*-dimension* η*.*

Proof. By Proposition 8, the set of exponent selections is the Cartesian product of: one word in $\{0,\dots,M\}^{a_O}$ for each compatible orbit O, and one exponent in $\{0,\dots,M\}$ for each distinct factor of each incompatible component. By Proposition 24, κ is the sum of the corresponding local contributions, and η_k is the sum of the compatible-orbit edge contributions (incompatible components contribute 0). A sum over a Cartesian product of a product of local monomials therefore factorizes as the product of the local sums. For a compatible orbit the local sum is $\mathscr{E}^{\circ(M)}_{a_O,w_O}(U,Z)$, which by Lemma 27 equals $\operatorname{tr}(T_{w_O}^{(M)}(U,Z)^{a_O})$. For an incompatible factor f, the local sum is $\sum_{e_f=0}^{M}U^{w_f(M-e_f)}=1+U^{w_f}+\cdots+U^{Mw_f}$ after substituting $r=M-e_f$. This proves (15); the coefficient interpretation is Definition 26 together with Proposition 8. □

At $(U,Z)=(1,1)$ every local factor counts its local selections, so Theorem 28 returns the checksum $\mathscr{E}^{\circ(M)}(1,1)=(M+1)^{N_F}$.

Remark 29 (What (15) does and does not require). The trace identity of Lemma 27 and the product formula (15) are identities in $\mathbb{Z}[U,Z]$ valid for every integer $M\geq 1$; that $M=p^t$ is a power of the characteristic is needed only for the field construction behind Propositions 8 and 24 (Lemma 5 and Proposition 16), not for the enumeration itself. Likewise the compatibility condition of Proposition 16 enters only through the split $\mathcal{S}_k^{\mathrm{comp}}\cup\mathcal{S}_k^{\mathrm{inc}}$: incompatible components are not excluded, they are counted by the dimension-only factors in (15). All coefficients count *labeled* codes; no quotient by monomial equivalence is taken. Finally, reversing the orientation of every orbit leaves η_k unchanged, since $\min\{M-a,b\}-\min\{M-b,a\}=b-a$ telescopes to 0 around a cycle; hence no statistic derived below depends on the choice of orientation. The predecessor convention in (11) does depend on it, and is fixed by the code-first pairing (7).

D. The simple-root case $M=1$

When $p\nmid n$ the modulus is square-free, $M=1$, every exponent is binary and the transfer matrix is 2×2; the general formula (15) then reduces to a completely explicit statement.

This is the classical simple-root regime, and we record it both as a check on the conventions and because it exhibits the combinatorics of the general case in its simplest form.

Proposition 30 (Simple-root specialization)**.** *Let $M = 1$ and $w \geq 1$. With rows and columns indexed by $\alpha, \beta \in \{0, 1\}$,*

$$T_w^{(1)}(U, Z) = \begin{pmatrix} U^w & 1 \\ U^w Z^w & 1 \end{pmatrix},$$

and the hull edge term of (11) *is*

$$\min\{e_i,\, 1 - e_{i+1}\} = \mathbf{1}[e_i = 1,\ e_{i+1} = 0],$$

the indicator of a descent $1 \to 0$ *along the orbit. Consequently*

$$\mathscr{E}^{(1)}(U, Z) = \prod_{s \in \mathcal{S}_k^{\mathrm{comp}}} \prod_{O \in \mathcal{O}_s} \mathrm{tr}\Big(T_{w_O}^{(1)}(U, Z)^{a_O}\Big) \times \prod_{s \in \mathcal{S}_k^{\mathrm{inc}}} \prod_{f \in \mathcal{F}_s} (1 + U^{w_f}).$$

and a compatible orbit contributes hull dimension zero exactly when its word is constant, so the LCD count is again $2^{\#\{\text{compatible orbits}\}}$ (times $2^{N_F^{\mathrm{inc}}}$, with N_F^{inc} as in Corollary 32, when some components are incompatible); the marginal κ is a sum of independent Bernoulli$(1/2)$ variables, weighted by w_f.

Proof. Put $M = 1$ in the definition of $T_w^{(M)}$. The entry in row α and column β is

$$U^{w(1-\beta)} Z^{w \min\{\alpha, 1-\beta\}},$$

which gives the four entries displayed. The edge term is $\min\{e_i, 1 - e_{i+1}\}$, which equals 1 iff $e_i = 1$ and $e_{i+1} = 0$ and vanishes otherwise. A cyclic binary word has no descent iff it is constant, so on an orbit of length a_O exactly 2 of the 2^{a_O} words give trivial hull, independently across orbits; the remaining statements are Theorem 28 and Corollary 32 specialized to $M = 1$ (both possible values of each exponent being equally weighted, κ is a weighted sum of independent fair bits). □

Example 31. Take $K = \mathbb{F}_4$, $n = 5$ (so $n' = 5$, $M = 1$, $p = 2 \nmid n$), $\lambda_0 = \lambda = 1$ and $k = 1$. The reciprocal fixes $x + 1$ and swaps the two quadratics, giving the orbits $(a_O, w_O) = (1, 1)$ and $(2, 2)$, hence $2^3 = 8$ simple-root cyclic codes with $W = 5$. With $T_2^{(1)} = \begin{pmatrix} U^2 & 1 \\ U^2 Z^2 & 1 \end{pmatrix}$ one

computes $\mathrm{tr}((T_2^{(1)})^2) = 1 + 2U^2Z^2 + U^4$, so

$$\mathscr{E}^{(1)}(U, Z) = (1 + U)(1 + 2U^2Z^2 + U^4).$$

The coefficients give $\mathbb{E}[\eta_1] = 1$ and $\mathrm{Var}(\eta_1) = 1$ (attained values of η_1 are 0 and 2, each for four codes), while the Z^0-part $(1+U)(1+U^4)$ shows that $2^2 = 4$ of the 8 codes are LCD. Both match Proposition 43, since $\nu_1 = 0$ (for $M = 1$, $\min\{e, 1 - e\} \equiv 0$) and $w_O^2 v(2, 1) = 4 \cdot \frac{1}{4} = 1$.

E. Immediate specializations and duality symmetry

Corollary 32. *Let* $N_F^{\mathrm{inc}} = \sum_{s \in \mathcal{S}_k^{\mathrm{inc}}} |\mathcal{F}_s|$. *Then*

(i) $\mathscr{E}^{(M)}(U, 1) = \prod_{s=1}^{N} \prod_{f \in \mathcal{F}_s} \big(1 + U^{w_f} + \cdots + U^{Mw_f}\big)$;

(ii) *the hull polynomial* $\mathscr{H}^{(M)}(Z) := \mathscr{E}^{(M)}(1, Z)$ *satisfies*

$$\mathscr{H}^{(M)}(Z) = (M + 1)^{N_F^{\mathrm{inc}}} \times \prod_{s \in \mathcal{S}_k^{\mathrm{comp}}} \prod_{O \in \mathcal{O}_s} \mathrm{tr}\Big(T_{w_O}^{(M)}(1, Z)^{a_O}\Big).$$

Proof. For (i), put $Z = 1$ in Definition 26. The remaining statistic κ is a sum over individual distinct factors by (12), so the sum factorizes factor by factor; the factor at f is $\sum_{e_f=0}^{M} U^{w_f(M-e_f)}$. Part (ii) is $U = 1$ in (15), where every incompatible factor becomes $M + 1$. □

Lemma 33. *For all* $a, b \in \{0, \ldots, M\}$,

$$\min\{a, M - b\} - \min\{M - a, b\} = a - b.$$

Proof. If $a + b \le M$, the two minima are a and b. If $a + b \ge M$, they are $M - b$ and $M - a$, whose difference is again $a - b$. □

Corollary 34. *For all* κ, η,

$$N_{\kappa,\eta} = N_{W-\kappa,\eta},$$

$$\textit{equivalently} \qquad \mathscr{E}^{(M)}(U, Z) = U^W \mathscr{E}^{(M)}(U^{-1}, Z).$$

as a Laurent polynomial identity in U.

Proof. Define a bijection on exponent selections coordinatewise as follows. On a compatible orbit set $e_i^\vee = M - e_{i-1}$ (the dual-generator exponents of Proposition 17); on an incompatible

factor set $e_f^\vee = M - e_f$. This is bijective because on each orbit it permutes the coordinates and complements them (it is not in general an involution: applying it twice shifts the indices by two). On a compatible orbit,

$$\sum_{i=1}^{a}(M - e_i^\vee) = \sum_{i=1}^{a} e_{i-1} = aM - \sum_{i=1}^{a}(M - e_i),$$

The edge sum of the transformed word is $\sum_i \min\{M - e_{i-1}, e_i\} = \sum_i \min\{M - e_i, e_{i+1}\}$ after re-indexing. By Lemma 33, its difference from $\sum_i \min\{e_i, M - e_{i+1}\}$ is $\sum_i (e_{i+1} - e_i) = 0$. On an incompatible factor, $w_f(M - e_f)$ is sent to $w_f M - w_f(M - e_f)$ and the hull contribution remains 0. Summing over all local pieces, the bijection sends $\kappa \mapsto W - \kappa$ and fixes η_k, proving the coefficient symmetry; the polynomial identity is a restatement. □

F. Short orbits

For orbit lengths 1 and 2 the trace can be evaluated completely.

Proposition 35. *For every $M \geq 1$ and $w \geq 1$:*

(i)
$$\mathscr{E}_{1,w}^{(M)}(U, Z) = \sum_{r=0}^{M} U^{wr} Z^{w \min\{r, M-r\}}.$$

In the hull polynomial $\mathscr{E}_{1,w}^{(M)}(1, Z)$, the coefficient of Z^{wh} is 2 for $0 \leq h < M/2$, is 1 for $h = M/2$ when M is even, and is 0 for $h > \lfloor M/2 \rfloor$.

(ii)
$$\mathscr{E}_{2,w}^{(M)}(U, Z) = \sum_{R=0}^{2M} \big(M + 1 - |R - M|\big)\, U^{wR} Z^{w \min\{R, 2M-R\}}.$$

Proof. (i) A length-one closed walk is a diagonal entry; with $r = M - e_1$ its monomial is $U^{wr} Z^{w \min\{M-r, r\}}$. The hull coefficients follow because $\min\{r, M - r\}$ takes each value $h < M/2$ twice (at $r = h$ and $r = M - h$) and the value $M/2$ once when M is even.

(ii) Let $s = e_1 + e_2$ and $R = 2M - s = \sum_i (M - e_i)$. If $s \leq M$, then $e_1 \leq M - e_2$ and $e_2 \leq M - e_1$, so the edge sum is s; if $s \geq M$ the reverse inequalities give $2M - s$. Hence the unweighted edge sum is $\min\{s, 2M - s\} = \min\{R, 2M - R\}$, a function of R alone. The number of pairs $(M - e_1, M - e_2) \in \{0, \ldots, M\}^2$ with sum R is $M + 1 - |R - M|$. □

Example 36. For $a = 2$ and $M = 2$, writing $R = \sum_i (2 - e_i)$ and $H = \sum_i \min\{e_i, 2 - e_{i+1}\}$, the nine words give

(e_1, e_2)	00	01	02	10	11	12	20	21	22
R	4	3	2	3	2	1	2	1	0
H	0	1	2	1	2	1	2	1	0

and therefore

$$\mathscr{E}^{\circ(2)}_{2,w}(U, Z) = 1 + U^{4w} + 2U^w Z^w + 2U^{3w} Z^w + 3U^{2w} Z^{2w},$$

in agreement with Proposition 35(ii) and visibly symmetric as in Corollary 34.

For $a \geq 3$ we do not assert a binomial-type closed form; the trace formula of Lemma 27 is the exact structural form used in this paper. Section VI turns it into an exact dynamic-programming computation and verifies it independently, while the support and the first two moments in Section V are closed.

V. Consequences

The marginals were already recorded in Corollary 32. This section extracts the structural consequences of Theorem 28: the LCD count (and its invariance in M when every component is compatible), the exact one-orbit and global supports, the mean, variance, covariance and conditional law of η_k under the uniform measure on exponent selections, and the counts of the four extremal classes.

A. The LCD count: invariance in M under compatibility

Proposition 37. *The number of k-Galois LCD codes is*

$$(M + 1)^{N_F^{\mathrm{inc}}} \, 2^{\sum_{s \in \mathcal{S}_k^{\mathrm{comp}}} |\mathcal{O}_s|}.$$

In particular, if every component is compatible, the number of LCD codes is $2^{\sum_s |\mathcal{O}_s|}$, independent of M.

Proof. On an incompatible component every code is LCD (Proposition 19), contributing the factor $(M + 1)^{|\mathcal{F}_s|}$. On one compatible orbit, $\eta_k = 0$ forces $\min\{e_i, M - e_{i+1}\} = 0$ for every i, i.e. $e_i = 0$ or $e_{i+1} = M$. If some $e_j > 0$ then $e_{j+1} = M$, then inductively every successor is M, and going once around the cycle forces the whole word to be $M \cdots M$. Otherwise the word is $0 \cdots 0$. Hence exactly two LCD words per compatible orbit; independence over orbits gives the formula. □

When all components are compatible, an LCD code has $\kappa = \sum_{O\in\mathcal{T}} a_O M w_O$ where $\mathcal{T}$ is the set of orbits carrying the word $0\cdots 0$; thus the $\eta = 0$ row of the joint table is supported on orbit subset-sums. Although the LCD *count* is then constant in M, the LCD *proportion* $2^{\#\text{orbits}}/(M+1)^{N_F}$ decays rapidly: repeated roots make LCD codes relatively rarer, not fewer.

B. One-orbit support: the trapezoid theorem

Fix one compatible orbit of length a and weight w, and write, for a word $\mathbf{e} = (e_1, \ldots, e_a)$,

$$R = \sum_{i=1}^{a} (M - e_i), \qquad S = aM - R = \sum_{i=1}^{a} e_i,$$

$$H = \sum_{i=1}^{a} \min\{e_i, M - e_{i+1}\}.$$

so its contribution to (κ, η_k) is (wR, wH). Termwise one has the upper bound $H \le \min\{R, S\}$; Theorem 39 identifies the exact support, including a lower bound that is invisible from dimensions alone, and shows that it is a trapezoid.

Lemma 38. *Let $a \ge 3$. If a word is changed by decrementing one coordinate $e_j > 0$ by 1 and incrementing the next coordinate $e_{j+1} < M$ by 1 (indices modulo a), then R and S are unchanged and H changes by -1, 0 or 1.*

Proof. Only the three edges $j-1 \to j$, $j \to j+1$, $j+1 \to j+2$ can change; they are distinct because $a \ge 3$. Writing $x = e_j$ and $y = e_{j+1}$, the middle term changes from $\min\{x, M-y\}$ to $\min\{x-1, M-y-1\} = \min\{x, M-y\} - 1$ (the minimum is ≥ 1 since $x \ge 1$ and $y \le M-1$). The incoming term $\min\{e_{j-1}, M-x\}$ increases by 0 or 1, and the outgoing term $\min\{y, M-e_{j+2}\}$ increases by 0 or 1. The total change is therefore in $\{-1, 0, 1\}$. □

Theorem 39. *For one orbit of length a, the attainable pairs (R, H) are exactly the integer pairs with*

$$0 \le R \le aM, \qquad \min\{R, M, aM - R\} \le H \le \min\{R, aM - R\}. \tag{16}$$

Equivalently the attainable contributions are (wR, wH) for such pairs.

The proof is given in Appendix A.

Corollary 40. *On one orbit, the values taken by H are exactly $0, 1, \ldots, \lfloor aM/2 \rfloor$. Globally, the set of values of η_k is the Minkowski sum*

$$\sum_{s\in\mathcal{S}_k^{\text{comp}}} \sum_{O\in\mathcal{O}_s} w_O \{0, 1, \ldots, \lfloor a_O M/2 \rfloor\}.$$

In particular, if all compatible orbit weights are equal to w, then η_k/w takes every integer value between 0 and $\sum_O \lfloor a_O M/2 \rfloor$.

Proof. No value exceeds $\lfloor aM/2 \rfloor$ since $H \le \min\{R, S\} \le aM/2$. Given $0 \le H \le \lfloor aM/2 \rfloor$, take $R = H$ in Theorem 39: then $S = aM - H \ge H$, so $\min\{R, S\} = H$ and H lies in the interval (16). The global statement is independence over orbits plus Proposition 24; a sum of integer intervals is an integer interval. □

Corollary 41. *A pair (κ, η) occurs iff, over the compatible orbits, one sums pairs $(w_O R_O, w_O H_O)$ from (16); to this one adds an incompatible contribution of the form $\big(\sum_{s \in \mathcal{S}_k^{\mathrm{inc}}} \sum_{f \in \mathcal{F}_s} w_f r_f,\, 0\big)$, where $0 \le r_f \le M$. In particular the necessary bound $0 \le \eta \le \min\{\kappa, W - \kappa\}$ from $\mathrm{Hull}_k(C) \subseteq C \cap D_k(C)$ is generally not sufficient: for $M \ge 2$, if a compatible orbit has $0 < R_O < a_O M$, its hull contribution is at least $w_O \min\{R_O, M, a_O M - R_O\} > 0$, whereas the endpoint choices $R_O = 0$ and $R_O = a_O M$ contribute 0; this creates the pointed gaps at small η visible in the computed tables of Section VI.*

C. Mean, variance, covariance and conditional law

The uniform measure on exponent selections makes the coordinates e_f independent uniform random variables on $\{0, \dots, M\}$, independent across orbits. For an edge with independent endpoints write $Y = \min\{X, M - Y'\}$ with X, Y' independent uniform on $\{0, \dots, M\}$.

Lemma 42. *For $1 \le r \le M$, $\Pr(Y \ge r) = \left(\frac{M-r+1}{M+1}\right)^2$, and*

$$\mu_M := \mathbb{E}[Y] = \frac{M(2M+1)}{6(M+1)}, \qquad \mathbb{E}[Y^2] = \frac{M(M^2+M+1)}{6(M+1)},$$

$$\mathrm{Var}(Y) = \frac{M(M+2)(2M^2+4M+3)}{36(M+1)^2}.$$

For two adjacent edges Y_i, Y_{i+1} in an orbit of length $a \ge 3$,

$$\mathbb{E}[Y_i Y_{i+1}] = \frac{M(M-1)(11M^2+15M+6)}{120(M+1)^2},$$

$$\mathrm{Cov}(Y_i, Y_{i+1}) = -\frac{M(M+2)(7M^2+14M+9)}{360(M+1)^2}.$$

For the two edges of an orbit of length $a = 2$, $\mathbb{E}[Y_1 Y_2] = \frac{M^2(M-1)}{12(M+1)}$.

Proof. The event $\{Y \geq r\}$ is $\{X \geq r,\ Y' \leq M - r\}$, of probability $\left(\frac{M-r+1}{M+1}\right)^2$. The layer-cake identities $\mathbb{E}[Y] = \sum_{r\geq 1} \Pr(Y \geq r)$ and $\mathbb{E}[Y^2] = \sum_{r\geq 1}(2r-1)\Pr(Y \geq r)$, with $\sum_{j=1}^{M} j^2 = M(M+1)(2M+1)/6$, give the first formulas. For adjacent edges in length $a \geq 3$, writing the shared variable second, $\{Y_i \geq r,\ Y_{i+1} \geq s\}$ requires the first variable $\geq r$, the third $\leq M - s$, and the shared variable in $[s, M - r]$; hence

$$\Pr(Y_i \geq r,\, Y_{i+1} \geq s) = \frac{(M-r+1)(M-s+1)}{(M+1)^3} \times (M-r-s+1)_+.$$

where $(t)_+ = \max\{t, 0\}$. Summing over $r, s \geq 1$ (a finite power-sum evaluation) gives $\mathbb{E}[Y_i Y_{i+1}]$, and subtracting μ_M^2 gives the covariance. For $a = 2$ the joint event $\{Y_1 \geq r,\ Y_2 \geq s\}$ says both variables lie in intervals of length $M - r - s + 1$, giving probability $(M-r-s+1)_+^2/(M+1)^2$ and the stated product expectation by the same summation. □

Define the loop mean and one-orbit variance functions

$$\nu_M = \frac{\lfloor M^2/4 \rfloor}{M+1} = \begin{cases} \dfrac{m^2}{2m+1}, & M = 2m, \\ \dfrac{m}{2}, & M = 2m+1, \end{cases}$$

and

$$v(a, M) = \begin{cases} v_1(M), & a = 1, \\ \dfrac{M(M+2)(M^2+2M+3)}{18(M+1)^2}, & a = 2, \\ a\,\dfrac{M(M+2)(M^2+2M+2)}{60(M+1)^2}, & a \geq 3, \end{cases}$$

where

$$v_1(M) = \begin{cases} \dfrac{M(M+2)(M^2+2M+4)}{48(M+1)^2}, & M = 2m, \\ \dfrac{(M-1)(M+3)}{48}, & M = 2m+1. \end{cases}$$

Proposition 43. *Under the uniform measure on exponent selections,*

$$\mathbb{E}[\eta_k] = \sum_{\substack{s \in \mathcal{S}_k^{\mathrm{comp}} \\ O \in \mathcal{O}_s,\, a_O = 1}} w_O \nu_M + \sum_{\substack{s \in \mathcal{S}_k^{\mathrm{comp}} \\ O \in \mathcal{O}_s,\, a_O \geq 2}} a_O w_O \mu_M,$$

$$\mathrm{Var}(\eta_k) = \sum_{s \in \mathcal{S}_k^{\mathrm{comp}}} \sum_{O \in \mathcal{O}_s} w_O^2\, v(a_O, M).$$

$$\mathbb{E}[\kappa] = \frac{W}{2},$$

$$\mathrm{Var}(\kappa) = \frac{M(M+2)}{12} \sum_{s=1}^{N} \sum_{f \in \mathcal{F}_s} w_f^2,$$

$$\mathrm{Cov}(\kappa, \eta_k) = 0.$$

Proof. By Proposition 24, η_k is a sum of independent compatible-orbit contributions. For $a \geq 2$ every edge has distinct endpoints, so each edge has mean μ_M. For $a = 1$ the single variable is $\min\{X, M-X\}$; Proposition 35(i) gives its distribution, whose mean and variance evaluate to ν_M and $v_1(M)$ by finite power sums (separating the parities of M). For the variance at $a = 2$, $\mathrm{Var}(Y_1 + Y_2) = 2\,\mathrm{Var}(Y) + 2(\mathbb{E}[Y_1 Y_2] - \mu_M^2)$ with Lemma 42 simplifies to $v(2, M)$. For $a \geq 3$ there are a unordered adjacent edge pairs on the cycle, and every non-adjacent pair (when $a \geq 4$) involves disjoint variables, so $\mathrm{Var}(\sum_i Y_i) = a\,\mathrm{Var}(Y) + 2a\,\mathrm{Cov}(Y_i, Y_{i+1}) = v(a, M)$; at $a = 3$ every pair is adjacent, which the same count encodes. Multiplying by w_O^2 and adding over independent orbits gives $\mathrm{Var}(\eta_k)$.

Each $M - e_f$ is uniform on $\{0, \ldots, M\}$, of mean $M/2$ and variance $((M+1)^2 - 1)/12 = M(M+2)/12$; this gives the κ moments via (6). Finally, Corollary 34 implies that, conditional on η_k, the variables κ and $W - \kappa$ have the same distribution; hence $\mathbb{E}[\kappa \mid \eta_k] = W/2$ and $\mathrm{Cov}(\kappa, \eta_k) = \mathbb{E}\big[\eta_k\, \mathbb{E}[\kappa - W/2 \mid \eta_k]\big] = 0$. □

Zero covariance here coexists with strong dependence: by Theorem 39 the conditional distribution of η_k given κ varies substantially with κ (see Example 54).

Proposition 44. *For $j = 0, 1, 2$ define the coefficients $\Phi_j(\kappa) = [U^\kappa]\left(\partial^j \mathscr{E}^{(M)}/\partial Z^j\right)(U, 1)$; for every attainable κ (that is, $\Phi_0(\kappa) > 0$) we have*

$$\Pr(\eta_k = \eta \mid \kappa) = \frac{N_{\kappa,\eta}}{\Phi_0(\kappa)}, \qquad \mathbb{E}[\eta_k \mid \kappa] = \frac{\Phi_1(\kappa)}{\Phi_0(\kappa)},$$

$$\mathrm{Var}(\eta_k \mid \kappa) = \frac{\Phi_2(\kappa) + \Phi_1(\kappa)}{\Phi_0(\kappa)} - \left(\frac{\Phi_1(\kappa)}{\Phi_0(\kappa)}\right)^2.$$

The conditional laws at κ and $W - \kappa$ coincide.

Proof. Extracting $[U^\kappa]$ from $\mathscr{E}^{(M)}$ and its Z-derivatives at $Z = 1$ gives $\Phi_0(\kappa) = \sum_\eta N_{\kappa,\eta}$, $\Phi_1(\kappa) = \sum_\eta \eta N_{\kappa,\eta}$ and $\Phi_2(\kappa) = \sum_\eta \eta(\eta-1) N_{\kappa,\eta}$; the formulas are the definitions of conditional probability, mean and variance. The last statement is Corollary 34. □

Remark 45. Fix a finite compatible orbit skeleton and let the integer M grow in the moment formulas. Since $\mu_M/M \to 1/3$, $\nu_M/M \to 1/4$ and $v(a,M)/M^2 \to 1/48$, $1/18$, $a/60$ in the three regimes, with $S_1 = \sum_{a_O=1} w_O$ and $S_{\geq 2} = \sum_{a_O \geq 2} a_O w_O$ (so $W = M(S_1 + S_{\geq 2})$),

$$\frac{\mathbb{E}[\eta_k]}{W} \longrightarrow \frac{\frac{1}{4}S_1 + \frac{1}{3}S_{\geq 2}}{S_1 + S_{\geq 2}} \in \left[\tfrac{1}{4}, \tfrac{1}{3}\right]:$$

a typical hull dimension interpolates between $W/4$ (length-1 orbits) and $W/3$ (longer orbits), with fluctuations of order M.

D. Extremal classes

Proposition 46. *For every global code C,*

$$C \subseteq D_{\mathcal{A},k}(C) \iff \eta_k(C) = \kappa(C),$$

$$D_{\mathcal{A},k}(C) \subseteq C \iff \eta_k(C) = W - \kappa(C),$$

$$C = D_{\mathcal{A},k}(C) \iff \eta_k(C) = \kappa(C) = \frac{W}{2}.$$

Proof. The hull is contained in both C and $D_{\mathcal{A},k}(C)$, and $\dim_{\mathbb{F}_q} D_{\mathcal{A},k}(C) = W - \kappa(C)$ by Proposition 10 applied componentwise. Equality of dimensions with a containing space forces equality of the spaces, giving all three criteria. □

The generator-level equivalents are what can be counted orbit by orbit. Let $J^{\geq}, J^{\leq}$ be the $(M+1)\times(M+1)$ zero-one matrices with $(J^{\geq})_{\alpha,\beta} = \mathbf{1}[\alpha+\beta \geq M]$ and $(J^{\leq})_{\alpha,\beta} = \mathbf{1}[\alpha+\beta \leq M]$, and put

$$\mathrm{sd}(a,M) = \begin{cases} M+1, & a \text{ even}, \\ 1, & a \text{ odd and } M \text{ even}, \\ 0, & a \text{ odd and } M \text{ odd}. \end{cases}$$

Proposition 47. *For a compatible component, $C_s(\mathbf{e})$ is self-orthogonal if and only if $e_{i-1}+e_i \geq M$ on* every *orbit and every cyclic index i; dual-containing if and only if $e_{i-1} + e_i \leq M$ throughout; and self-dual if and only if $e_{i-1} + e_i = M$ throughout. The numbers of words satisfying the respective condition on one orbit of length a are $\mathrm{tr}((J^{\geq})^a)$, $\mathrm{tr}((J^{\leq})^a)$ and $\mathrm{sd}(a,M)$ respectively; moreover $\mathrm{tr}((J^{\geq})^a) = \mathrm{tr}((J^{\leq})^a)$, equal to $\lfloor M/2 \rfloor + 1$ for $a = 1$ and to $(M+1)(M+2)/2$ for $a = 2$. Globally, the numbers of self-orthogonal and dual-containing*

codes are

$$\prod_{s\in\mathcal{S}_k^{\mathrm{comp}}}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\big((J^{\geq})^{a_O}\big),$$

and

$$\prod_{s\in\mathcal{S}_k^{\mathrm{comp}}}\prod_{O\in\mathcal{O}_s}\operatorname{tr}\big((J^{\leq})^{a_O}\big).$$

(incompatible components contribute a factor 1 *by Corollary 20), while the number of self-dual codes is* 0 *if* $\mathcal{S}_k^{\mathrm{inc}}\neq\varnothing$*, and otherwise* $\prod_{s,O}\mathrm{sd}(a_O,M)$*.*

Proof. On a compatible component, $C_s(\mathbf{e})=\langle g_{\mathbf{e}}\rangle$ and $D_k(C_s(\mathbf{e}))=\langle h_{\mathbf{e}}^{\#_{s,k}}\rangle$ in the same quotient. Inclusion of principal ideals reverses divisibility of the monic generators: $C_s(\mathbf{e})\subseteq D_k(C_s(\mathbf{e}))$ iff $h_{\mathbf{e}}^{\#_{s,k}}\mid g_{\mathbf{e}}$, which by Proposition 17 says $M-e_{i-1}\leq e_i$ at every factor, i.e. $e_{i-1}+e_i\geq M$. The other inclusion and equality are identical. Counting cyclic words by allowed adjacent pairs is counting closed walks, hence the two traces. The complement map $e_i\mapsto M-e_i$ exchanges the two inequalities, proving equality of the traces; the values at $a=1$ (the condition is $2e_1\geq M$, resp. $\leq M$) and at $a=2$ (the condition is on the single sum e_1+e_2) are direct counts.

For self-duality on one orbit, $e_i=M-e_{i-1}$ forces the word to alternate $(\alpha,M-\alpha,\alpha,\ldots)$. If a is even any $\alpha\in\{0,\ldots,M\}$ works; if a is odd, closing the cycle forces $\alpha=M-\alpha$, possible only for M even, with $\alpha=M/2$. This is $\mathrm{sd}(a,M)$.

Globally the three properties hold iff they hold in every component. By Corollary 20 an incompatible component admits exactly one self-orthogonal code ($\{0\}$), exactly one dual-containing code (K_s^n) and no self-dual code, which completes the formulas. □

Corollary 48. *A self-dual code exists if and only if every component is compatible and no orbit has* a_O *odd with* M *odd. When* $p=2$ *and* $t\geq 1$*, the multiplicity* $M=p^t$ *is even, so every fully compatible orbit skeleton admits self-dual codes.*

VI. Computation and examples

This section describes how the coefficients of $\mathscr{E}^{(M)}$ are computed, states the complexity of the dynamic program, gives worked examples in characteristics 2, 3 and 5, and explains the independent verification. The preprocessing step, i.e. the computation of the orbit data from the factorization of the kernel, is recorded in Appendix B. Every number below is generated by the supplementary code and checked against its output files.

A. The enumerator dynamic program and its complexity

The primary implementation evaluates the trace in Theorem 28 literally. For one compatible orbit it keeps, for every pair (start state, current state) in $\{0,\dots,M\}^2$, a sparse dictionary of accumulated (κ,η) counts (multiplied into the running global polynomial), extends the walk edge by edge, and after a_O steps keeps the diagonal start = current. The only mathematical line is the transition

$$(\kappa,\eta)\longmapsto\big(\kappa+w(M-\beta),\ \eta+w\min\{\alpha,M-\beta\}\big)\qquad(\alpha\to\beta),$$

which is (14). An incompatible factor is convolved with $1+U^{w_f}+\cdots+U^{Mw_f}$. In the reference implementation each one-orbit polynomial is computed separately (caching repeated pairs (a_O,w_O)) and the orbit polynomials are then convolved; every one-orbit polynomial is in addition cross-checked against brute-force enumeration of its words in §VI-C. All arithmetic on coefficients and moments is exact (`int` and `fractions.Fraction` in the Python reference implementation); floating point occurs only in plots and in explicitly marked decimal approximations.

Proposition 49. *Given the compatible orbit data and the incompatible factor weights, a dense implementation of the closed-walk dynamic program computes all coefficients of $\mathscr{E}^{(M)}$ in*

$$O\Big(N_F(M+1)^3\,(W+1)\big(\lfloor W/2\rfloor+1\big)\Big)$$

exact-integer additions and $O\big((M+1)^2(W+1)(\lfloor W/2\rfloor+1)\big)$ stored coefficients. The coefficients have $O(N_F\log(M+1))$ bits.

Proof. Every intermediate pair satisfies $0\le\kappa\le W$ and $0\le\eta\le\lfloor W/2\rfloor$ (the final hull satisfies $\eta\le\min\{\kappa,W-\kappa\}\le\lfloor W/2\rfloor$, and partial sums only increase), so every dictionary is supported in a rectangle with $(W+1)(\lfloor W/2\rfloor+1)$ cells. Let $N_F^{\mathrm{comp}}=\sum_{s\in\mathcal{S}_k^{\mathrm{comp}}}|\mathcal{F}_s|$ be the number of distinct kernel factors on compatible components; this is also the total number of compatible edge-steps, and $N_F^{\mathrm{comp}}\le N_F$. At each step there are $(M+1)^2$ (start, current) state pairs and $M+1$ choices of next state, each requiring one pass over the rectangle. The incompatible convolutions are dominated by the same bound. A coefficient counts a subset of the $(M+1)^{N_F}$ selections, hence has $O(N_F\log(M+1))$ bits. □

Proposition 49 is an arithmetic-operation bound for the dense worst case; polynomial factorization and orbit construction are excluded from this bound. The sparse reference implementation is well below the dense bound in practice: for $\mathbb{F}_{32}$, $M=4$, $W=124$ the enumerator has only 3531 nonzero coefficients (Example 54). Obvious accelerations (not

needed here) are dividing U, Z exponents by $\gcd_O w_O$, powering cached one-orbit polynomials for repeated orbits, and prefix-sum evaluation of the min-transitions.

B. Worked examples

In Examples 50–54 the algebra $\mathcal{A}$ is a single field K (so $N = 1$, $m_s = 1$, $q = |K|$), except in Example 52. We always record λ_0 and $\lambda = \lambda_0^M$ (Remark 6).

Example 50. Let $K = \mathbb{F}_4 = \mathbb{F}_2(a)$, $a^2 + a + 1 = 0$, $n = 10$, so $n' = 5$, $M = 2$, and take $\lambda_0 = \lambda = 1$, $k = 1$. Then

$$x^5 - 1 = (x + 1)(x^2 + ax + 1)(x^2 + (a + 1)x + 1),$$

the reciprocal fixes $x + 1$ and swaps the two quadratics, giving orbits $(a_O, w_O) = (1, 1), (2, 2)$ and $3^3 = 27$ codes. The joint multiplicities $N_{\kappa,\eta}$ are:

$\kappa\backslash\eta$	0	1	2	3	4	5
0	1	–	–	–	–	–
1	–	1	–	–	–	–
2	1	–	2	–	–	–
3	–	–	–	2	–	–
4	–	–	2	–	3	–
5	–	–	–	–	–	3
6	–	–	2	–	3	–
7	–	–	–	2	–	–
8	1	–	2	–	–	–
9	–	1	–	–	–	–
10	1	–	–	–	–	–

Reading the table: the 4 LCD codes sit at $\kappa \in \{0, 2, 8, 10\}$, the orbit subset-sums of $\{a_O M w_O\} = \{2, 8\}$; the 3 codes at $(\kappa, \eta) = (5, 5)$ are exactly the self-dual ones (Proposition 46); and $\mathbb{E}[\eta_1] = 23/9$, $\mathrm{Var}(\eta_1) = 194/81$. The symmetry $\kappa \leftrightarrow 10 - \kappa$ is Corollary 34. The same orbit data and table arise for $\lambda_0 = \omega$ (then $\lambda = \omega^2$), as Table II records.

Example 51. Let $K = \mathbb{F}_8$, $n = 14$ ($n' = 7$, $M = 2$), $\lambda_0 = \lambda = 1$, $k = 1$. The orbits are $(1, 1)$ and $(6, 1)$, giving $3^7 = 2187$ codes and 44 nonzero joint coefficients. The LCD count is 4 (Proposition 37); moreover $\mathbb{E}[\eta_1] = 11/3$, $\mathrm{Var}(\eta_1) = 10/9$, and $\mathbb{E}[\eta_1 \mid \kappa = 7] = 533/131 \approx 4.069$. The joint distribution is Figure 2; the missing cells at $\eta = 1$ are a visible consequence of Theorem 39.

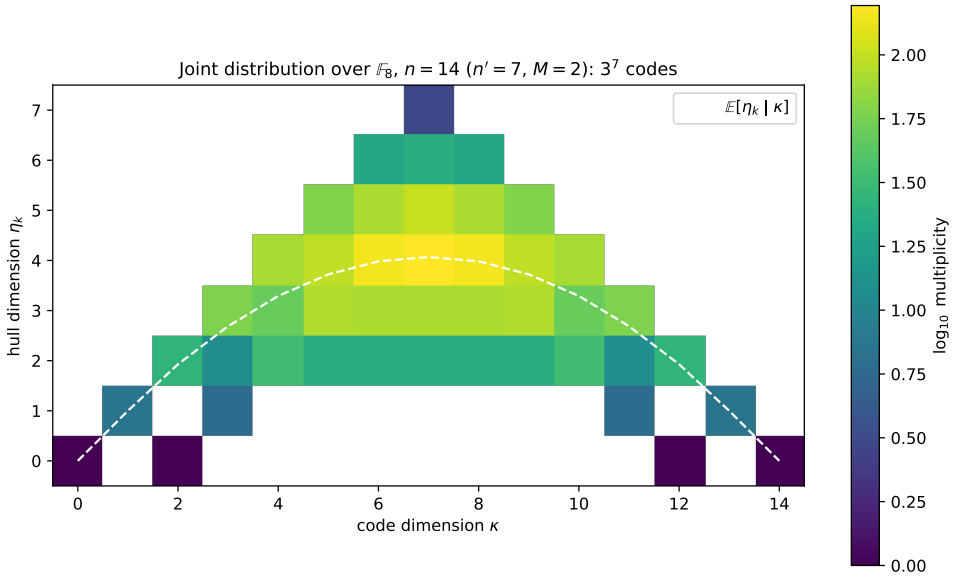


Fig. 2. Joint distribution $N_{\kappa,\eta}$ for $\mathbb{F}_8$, $n = 14$ (Example 51); color is $\log_{10} N_{\kappa,\eta}$ and white cells are zero. Only $\kappa \in \{1, 3, 11, 13\}$ occur at $\eta = 1$—the one-orbit trapezoid gaps of Theorem 39, invisible from the hull marginal alone.

Example 52. Let $\mathcal{A} = \mathbb{F}_4 \times \mathbb{F}_{16}$ over $\mathbb{F}_4$, $n = 10$ ($n' = 5$, $M = 2$), $\lambda_0 = \lambda = (1, 1)$, $k = 1$. The $\mathbb{F}_4$ component contributes orbits $(1, 1), (2, 2)$ as in Example 50; the $\mathbb{F}_{16}$ component has $m_s = [\mathbb{F}_{16} : \mathbb{F}_4] = 2$ and contributes $(1, 2), (4, 2)$—the weight $w_O = m_s d_O$ doubling in action. Here $W = 30$, there are $3^8 = 6561$ codes, 123 joint terms, 16 LCD codes, 9 self-dual codes, and $\mathbb{E}[\eta_1] = 23/3$, $\mathrm{Var}(\eta_1) = 458/81$. This family is obtained from the $\mathbb{F}_4$ skeleton of Example 50 and the $\mathbb{F}_{16}$ skeleton with the weights of Proposition 8 doubled, and is reproduced by the enumerator (Table I).

Example 53. Three families in odd characteristic illustrate the $a_O = 1$ and $a_O = 2$ phenomena of Sections IV-F and V.

(a) $K = \mathbb{F}_3$, $n = 12$ ($n' = 4$, $M = 3$), $\lambda_0 = \lambda = 1$, $k = 0$: orbits $(1, 1), (1, 1), (1, 2)$; 64 codes, 33 joint terms, 8 LCD codes, and *no* self-dual code—every orbit has odd length and M is odd (Corollary 48).

(b) $K = \mathbb{F}_9 = \mathbb{F}_3(i)$, $n = 12$ ($n' = 4$, $M = 3$), $\lambda_0 = i$, $\lambda = -i$, $k = 1$: a single orbit $(2, 2)$, only 16 codes and 7 joint terms. By Proposition 35(ii) the hull dimension is forced: $\eta = \min\{\kappa, 12 - \kappa\}$; there are 4 self-dual codes.

(c) $K = \mathbb{F}_{25}$, $n = 30$ ($n' = 6$, $M = 5$), $\lambda_0 = \lambda = 1$, $k = 1$: six fixed factors, orbits $(1, 1)^6$;

46656 codes, and all hull contribution comes from length-1 orbits through $\min\{e, 5-e\}$, with $\mathbb{E}[\eta_1] = 6$—note the loop mean $\nu_5 = 1$, not μ_5 (Proposition 43).

Example 54. Let $K = \mathbb{F}_{32}$, $n' = 31$, $\lambda_0 = \lambda = 1$, $k = 1$. By (17) the reciprocal acts on the 31 root exponents by $j \mapsto -16j \bmod 31$, of order 10 on the nonzero residues; hence the orbit skeleton is $(1, 1), (10, 1), (10, 1), (10, 1)$ for every $M = 2^t$.

- $M = 2$, $n = 62$: $3^{31} = 617{,}673{,}396{,}283{,}947$ codes, 916 nonzero joint coefficients, LCD 16, self-orthogonal 70,639,674,082, self-dual 27, $\mathbb{E}[\eta_1] = 17$, $\mathrm{Var}(\eta_1) = 14/3$, and $\max_\kappa \mathbb{E}[\eta_1 \mid \kappa] \approx 17.392$ at $\kappa = 31$.
- For $M = 4$ and $n = 124$ (so $t = 2$), there are $5^{31} = 4{,}656{,}612{,}873{,}077{,}392{,}578{,}125$ codes and 3531 joint coefficients; LCD is still 16, self-dual codes number $125 = 5^3$, $\mathbb{E}[\eta_1] = 184/5$ and $\mathrm{Var}(\eta_1) = 326/25$.

In both cases the exact joint distribution over all $(M+1)^{31}$ codes is obtained from the four orbit traces, with no enumeration of codes. Figure 3 shows the $M = 2$ distribution. For $M = 2, 4$, the conditional means $\mathbb{E}[\eta_1 \mid \kappa]$ peak at about $0.281\,W$ and $0.302\,W$, in line with the $W/3$ regime of Remark 45 (the skeleton limit is $\frac{1/4+30/3}{31} \approx 0.331$).

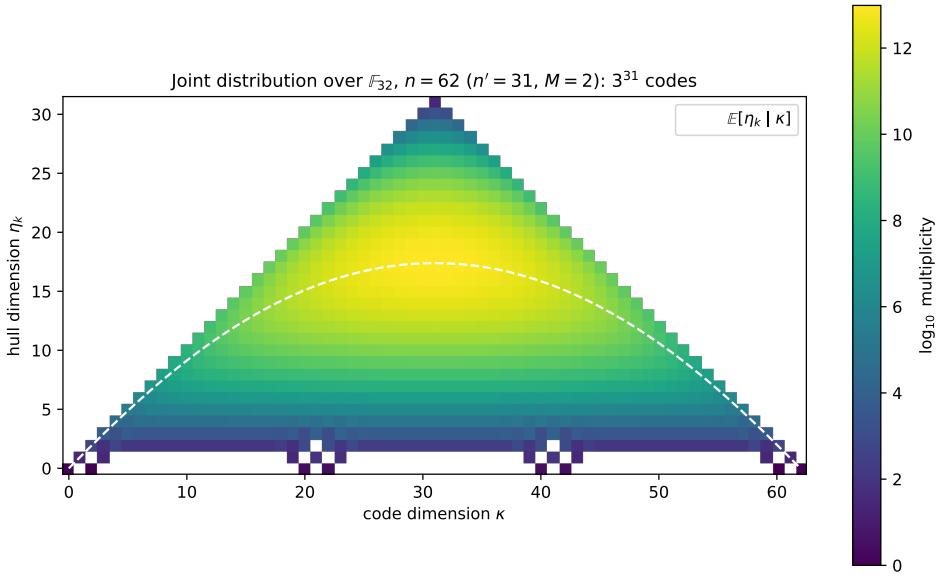


Fig. 3. Exact joint distribution for $\mathbb{F}_{32}$, $n = 62$, $M = 2$ (Example 54), over all 3^{31} codes; color is $\log_{10} N_{\kappa,\eta}$. The periodic white holes at small η are the superposition of the three length-10 orbit trapezoids.

TABLE I

COMPUTATIONAL RESULTS FOR THE CONSTRUCTED CODE FAMILIES AND THEIR ORBIT STRUCTURES, WITH CORRESPONDING LCD AND SD COUNTS AND STATISTICAL PROPERTIES OF THE k-GALOIS HULL DIMENSION.

Family	(n', M, n)	k	orbits (a_O, w_O)	codes	terms	LCD	SD	$\mathbb{E}\eta_k$	$\operatorname{Var}\eta_k$
$\mathbb{F}_4$, $\lambda_0{=}1$	$(5,2,10)$	1	$(1,1),(2,2)$	27	15	4	3	23/9	194/81
$\mathbb{F}_4$, $\lambda_0{=}\omega$	$(5,2,10)$	1	$(1,1),(2,2)$	27	15	4	3	23/9	194/81
$\mathbb{F}_8$	$(7,2,14)$	1	$(1,1),(6,1)$	2187	44	4	3	11/3	10/9
$\mathbb{F}_3$	$(4,3,12)$	0	$(1,1)^2,(1,2)$	64	33	8	0	2	3/2
$\mathbb{F}_9$, $\lambda_0{=}i$	$(4,3,12)$	1	$(2,2)$	16	7	2	4	7/2	15/4
$\mathbb{F}_9$, $\lambda_0{=}-1$	$(4,3,12)$	1	$(2,1)^2$	256	22	4	16	7/2	15/8
$\mathbb{F}_9$	$(8,3,24)$	1	$(1,1)^4,(2,1)^2$	65536	142	64	0	11/2	23/8
$\mathbb{F}_{25}$	$(6,5,30)$	1	$(1,1)^6$	46656	191	64	0	6	4
$\mathbb{F}_4{\times}\mathbb{F}_{16}$	$(5,2,10)$	1	$(1,1),(2,2),(1,2),(4,2)$	6561	123	16	9	23/3	458/81
$\mathbb{F}_{32}$	$(31,2,62)$	1	$(1,1),(10,1)^3$	3^{31}	916	16	27	17	14/3
$\mathbb{F}_{32}$	$(31,4,124)$	1	$(1,1),(10,1)^3$	5^{31}	3531	16	125	184/5	326/25

C. Independent verification

The computations are checked at three independent levels. The exact root-level command `python3 -I 04_supplementary/code/checks.py` prints a pass line for every check and ends in `ALL CHECKS PASSED`; its saved output is `checks_output.txt`. All shipped verification code is standard-library Python.

Level 1: internal combinatorics. The script `checks.py` compares the transfer-matrix trace with brute-force enumeration of all exponent words for every (a, M, w) with $a \le 6$, $M \le 5$ and $w \in \{1, 2\}$ (60 cases); checks the closed forms of Proposition 35; verifies the complement–reversal symmetry, the checksum $\mathscr{E}^{\circ}(1,1) = (M+1)^{N_F}$, the LCD count and the one-orbit trapezoid of Theorem 39 for $M \le 5$, $a \le 6$; verifies the moment formulas of Proposition 43 for $M \le 7$ on a range of skeletons; and recomputes every row of Table I, the joint table of Example 50, the conditional mean of Example 51 and the maximal conditional mean of Example 54 from the enumerator alone. The moment, covariance and complexity formulas are additionally re-derived symbolically in the external report `symbolic_verification.txt`; no shipped file imports a computer algebra system.

Level 2: direct finite-field linear algebra. For the families of Table II the script starts from the definitions, with no orbit input: it factors $x^{n'} - \lambda_{s,0}$ over K_s into distinct irreducible factors using the genuine irreducibility test of Appendix B, computes the reciprocal $f^{\#_{s,k}}$ from its definition (8), and reads off the orbit skeleton, which is found to agree with the

TABLE II

Exhaustive Computational Verification of the Trace-Product Formula

Family	p^m	n'	M	n	k	orbits	words checked
$\mathbb{F}_4$, $\lambda_0 = 1$	2^2	3	2	6	1	$(1,1)^3$	$3^3 = 27$
$\mathbb{F}_4$, $\lambda_0 = 1$	2^2	5	2	10	1	$(1,1),(2,2)$	$3^3 = 27$
$\mathbb{F}_4$, $\lambda_0 = \omega$	2^2	5	2	10	1	$(1,1),(2,2)$	$3^3 = 27$
$\mathbb{F}_8$, $\lambda_0 = 1$	2^3	7	2	14	1	$(1,1),(6,1)$	$3^7 = 2187$
$\mathbb{F}_3$, $\lambda_0 = 1$	3^1	4	3	12	0	$(1,1)^2,(1,2)$	$4^3 = 64$
$\mathbb{F}_9$, $\lambda_0 = i$	3^2	4	3	12	1	$(2,2)$	$4^2 = 16$
$\mathbb{F}_9$, $\lambda_0 = -1$	3^2	4	3	12	1	$(2,1)^2$	$4^4 = 256$

skeletons used in Examples 50–53. For *every* exponent word it then builds $g_{\mathbf{e}} = \prod f^{e_f}$, forms the generator matrix B from the $n - \deg g_{\mathbf{e}}$ non-wrapped shifts of $g_{\mathbf{e}}$, and computes

$$\eta_k = \dim C - \operatorname{rank}\big(B\,\sigma_{s,k}(B)^{\mathsf{T}}\big)$$

over K_s (Remark 22). The resulting joint histogram is compared with the trace product of Theorem 28; every word and every histogram matches. Characteristics 2 and 3, the Galois indices $k = 0$ and $k = 1$, the multiplicities $M = 2$ and $M = 3$, and both fixed and swapped factors are represented, so that the sign and normalization choices in $f^{\#_{s,k}}$ are exercised outside a single degenerate situation.

Level 3: independent brute-force families. To test the theorems outside the skeletons of Tables I–II, the same script constructs 21 families directly from the definitions: $\mathbb{F}_2$ and $\mathbb{F}_4$ with $n = 4, 6, 12$; $\mathbb{F}_3$ with $n = 6, 9$; $\mathbb{F}_5$ with $n = 10$; an $\mathbb{F}_{64}$ component over the base field $\mathbb{F}_8$ with $k = 1$, $\lambda_{s,0}$ of order 3 and $\gcd(\operatorname{ord}\lambda_{s,0}, n') \neq 1$; $k = 0$ and $k = 1$; compatible and incompatible twists with $\lambda_{s,0} \neq 1$, including the $\mathbb{F}_5$ example above; and the product algebras $\mathbb{F}_4 \times \mathbb{F}_{16}$ and $\mathbb{F}_4 \times \mathbb{F}_4$. For each family it checks the compatibility condition $\lambda_{s,0}^{1+p^{d_s-k}} = 1$, the reciprocal permutation, the Gram-rank hull dimension against the min-edge formula, the full joint histogram, W, moments, covariance, LCD/self-orthogonal/self-dual counts and every conditional mean against the enumerator. These checks include the $\mathbb{F}_4 \times \mathbb{F}_{16}$ row of Table I and the incompatible-twist example, and they remove the earlier limitation that larger families were only checked at the transfer-product level. The Level-3 families include $N_F^{\mathrm{inc}} > 0$ as well as fully compatible cases, so the LCD formula $(M+1)^{N_F^{\mathrm{inc}}} 2^{\#\mathrm{orbits}}$, the self-orthogonal count and the self-dual count are tested with incompatible components present. The saved output was regenerated from the exact isolated-mode command above; it is `checks_output.txt` in

the supplementary material.

VII. Conclusion

In this paper, we have determined the exact joint distribution of code dimension and k-Galois hull dimension for all $\mathcal{A}$-linear λ-constacyclic codes of arbitrary length over a square-free affine algebra. Writing each repeated-root modulus as a power of its square-free kernel reduces a code to an exponent selection; on each compatible reciprocal orbit the hull statistic is a cyclic sum of pairwise minima and is encoded by a transfer-matrix trace, while incompatible components contribute explicit dimension-only factors. The resulting enumerator yields the marginal laws, exact one-orbit support, means, variances and conditional laws, and the numbers of LCD, self-orthogonal, dual-containing and self-dual codes. Given the orbit data, its coefficients can be computed in time polynomial in n, M and the number of distinct kernel factors; the implementation has been checked by direct finite-field linear algebra and evaluates, in particular, the enumerator for all 3^{31} codes of length 62 over $\mathbb{F}_{32}$. Throughout, codes are counted as labeled $\mathcal{A}$-submodules, and no minimum-distance assertion is made.

The scope of the result should be stated as plainly as its content. First, all counts are over *labeled* exponent selections: codes are not identified up to monomial equivalence or isometry, so the enumerator describes the raw population of $\mathcal{A}$-submodules rather than a set of equivalence classes. Second, the formulas presuppose the orbit data of the reciprocal permutation $\tau_{s,k}$ on $\mathcal{F}_s$; producing that data requires factoring the square-free kernel $x^{n'} - \lambda_{s,0}$ over K_s, which is inexpensive for the lengths treated here but is a genuine input rather than something the enumerator supplies. Third, the parameter $M = p^t$ is common to all components because the length n is; different lengths over different components are not covered. Fourth, nothing here concerns the minimum distance, the weight enumerator, or the decoding problem, and no claim of optimality for the codes counted is made.

Beyond these limits, two directions seem most natural. One is the joint law of the code dimension κ together with the hull dimensions $\eta_{k_1}, \ldots, \eta_{k_r}$ for several Galois parameters $k_1, \ldots, k_r$; such a multivariate enumerator would require a transfer-state description that tracks the exponent relations arising from all the reciprocal actions simultaneously, and it is not a consequence of the single-k formula. The other is the passage from labeled to unlabeled counts by combining the transfer-matrix traces with a Burnside average over the monomial automorphism group of the ambient algebra.

Appendix A
Proof of Theorem 39

Proof: Recall that the pair attached to a word $\mathbf{e} = (e_1, \ldots, e_a)$ is (wR, wH) with $R = \sum_i (M - e_i)$, $S = aM - R$ and $H = \sum_i \min\{e_i, M - e_{i+1}\}$; the statement to be proved is that the attainable pairs (R, H) are exactly those satisfying (16).

Necessity. The upper bound is termwise: $\min\{e_i, M - e_{i+1}\} \le e_i$ gives $H \le S$, and $\min\{e_i, M - e_{i+1}\} \le M - e_{i+1}$ gives $H \le R$.

For the lower bound, first suppose $\min\{R, S\} \le M$. If $R \le S$ and $R \le M$, put $x_i = M - e_i$. For $a \ge 2$ the i-th edge term is $\min\{M - x_i, x_{i+1}\}$; were it $< x_{i+1}$, then $x_i + x_{i+1} > M \ge R$, impossible since x_i and x_{i+1} are distinct summands of R (for $a = 1$, $H = \min\{R, S\}$ directly). Hence every term equals x_{i+1} and $H = R$. Symmetrically, if $S \le R$ and $S \le M$ then $H = S$. Thus $H = \min\{R, S\} = \min\{R, M, S\}$ in this case.

It remains to show $H \ge M$ when $R, S \ge M$. The function $(u, v) \mapsto \min\{u, M - v\}$ is a minimum of two affine functions, hence concave, so H is a concave function on the polytope $\{0 \le e_i \le M,\ \sum e_i = S\}$. At a vertex of this polytope at most one coordinate lies in $(0, M)$ (else two such coordinates could be perturbed by $\pm\varepsilon$). At such a vertex, $S \ge M$ forces at least one coordinate equal to M and $R \ge M$ forces at least one coordinate equal to 0. Traveling along the cycle from a coordinate equal to M to the next coordinate equal to 0, one meets either a direct edge $M \to 0$, contributing M to H, or a passage $M \to c \to 0$ through the unique intermediate coordinate c (possibly preceded by zero-contribution edges $M \to M$), contributing $(M - c) + c = M$. Hence $H \ge M$ at every vertex; since every point of the polytope is a convex combination of vertices, concavity gives $H(\mathbf{e}) \ge M$ everywhere.

Attainability. For $a \le 2$ the interval in (16) is a single point (Proposition 35), so let $a \ge 3$. If $\min\{R, S\} \le M$ the interval is again a single point and necessity shows that every word with this R attains it. Assume therefore $R, S \ge M$, so the interval is $[M, \min\{R, S\}]$.

Lower endpoint. Write $S = \alpha M + c$ with $1 \le \alpha \le a - 1$ and $0 \le c < M$, and take the block word $(M^\alpha, c, 0^{a-\alpha-1})$, retaining c as a coordinate also when $c = 0$. Its only possible nonzero edge contributions are $M - c$ and c (the second being 0 when $c = 0$), so $H = M$.

Upper endpoint. Write $S = aq + r$, $0 \le r < a$, and start with r entries equal to $q + 1$ and $a - r$ entries equal to q, to be arranged cyclically. If $S \le R$ then $S \le aM/2$ and we want adjacent sums $\le M$, which gives $\min\{e_i, M - e_{i+1}\} = e_i$ and $H = S$. The only adjacent pair that could exceed M is $(q + 1, q + 1)$ with $2q + 2 > M$. If $r \ge 2$ and $2q + 2 > M$, then $S = aq + r \le aM/2$ forces M odd and $q = (M - 1)/2$: for even M, $2q + 2 > M$ gives $q \ge M/2$,

and with $r \geq 2$ one has $S = aq + r > aM/2$, a contradiction. Moreover $r = S - aq \leq a/2$, so the r entries $q+1$ can be placed with no two cyclically adjacent. In all other cases every adjacent sum is at most M automatically. If $S \geq R$ we instead want adjacent sums $\geq M$, giving $\min\{e_i, M - e_{i+1}\} = M - e_{i+1}$ and $H = R$. The only possible bad pair is (q, q) with $2q < M$; if $a - r \geq 2$ and $2q < M$, then $S \geq aM/2$ forces M odd and $q = (M-1)/2$: for even M, $2q < M$ gives $q \leq (M-2)/2$ and then $S = aq + r < aM/2$. In the odd case $r \geq a/2$, so the $a - r$ entries q can be placed with no two cyclically adjacent. Otherwise every adjacent sum is at least M automatically. In both cases $H = \min\{R, S\}$.

Filling the interval. Starting from the balanced upper-endpoint word, repeatedly move one unit from position $i+1$ to position i (linear order $1, \ldots, a$) whenever $e_i < M$ and $e_{i+1} > 0$. Each move increases $\sum_i (a-i)e_i$ by 1, so the process stops; it stops exactly at a word of the form $M, \ldots, M, c, 0, \ldots, 0$, which by the fixed sum S is the block word above. Each move is the reverse of the transfer in Lemma 38, so H changes by at most 1 per move. Along this path H travels from $\min\{R, S\}$ to M, hence attains every integer between them.

■

Appendix B
Computing the Orbit Data

The input is (p, e, k, n) together with the components (K_s, λ_s). For each component:

1) compute $t = v_p(n)$, $M = p^t$, $n' = n/M$ and $\lambda_{s,0} = \lambda_s^{p^{d_s - t \bmod d_s}} = \mathrm{Frob}^{-t}(\lambda_s)$;
2) factor the square-free polynomial $P_{s,0} = x^{n'} - \lambda_{s,0}$ over K_s into distinct monic irreducibles $\mathcal{F}_s$;
3) for each $f \in \mathcal{F}_s$, compute $f^{\#_{s,k}}$ by (8) and locate it in $\mathcal{F}_s$. Success for every f is exactly compatibility (Proposition 16), and the matching is $\tau_{s,k}$; failure identifies an incompatible component, for which only the list $\{w_f\}$ is retained;
4) decompose $\tau_{s,k}$ into cycles and record (a_O, w_O).

Step 2 can be carried out in any computer algebra system. For the fields used in Section VI-C, the verification program is written in plain Python and uses no external algebra package. It implements K_s directly, extracts all linear factors by evaluating $P_{s,0}$ at every field element, then recursively splits the remaining polynomial by trial division by every monic polynomial of degree at most half its current degree. A polynomial with no such divisor is irreducible; this is a genuine irreducibility test, not an inference from the size of a cofactor. Thus the

pipeline is self-contained, takes no orbit data as input, and also handles the extension-field and product-algebra cases of Level 3 below.

When $\lambda_{s,0} = 1$ there is a root-exponent shortcut. Let α be a primitive n'-th root of unity in an algebraic closure. The irreducible factors of $x^{n'} - 1$ over K_s correspond to the orbits of multiplication by $|K_s|$ on $\mathbb{Z}/n'\mathbb{Z}$ (the K_s-cyclotomic cosets), and by Lemma 13 the reciprocal acts on root exponents by

$$j \longmapsto -p^{d_s - k} j \pmod{n'}. \tag{17}$$

This is how the $\mathbb{F}_{32}$ orbit skeleton in Example 54 is obtained without writing down 31 linear factors.

Acknowledgment

All symbolic and numerical checks reported in Section VI-C were carried out with the two Python scripts `enumerator.py` and `checks.py` accompanying the manuscript; the saved output of the latter reproduces every table and every example of the paper. The authors used the AI-assisted writing tool Arena.ai Agent Mode at the level of literature organization in Section I, language and structural editing throughout the article, development and checking of the supplementary Python code used in Section VII, and assistance with the mathematical review and source-level checks reported in Section VII and Appendices A–B. The tool was not used to generate mathematical results, proofs, data, figures or tables independently, and it is not an author. All mathematical statements, proofs and computed data were reviewed and verified by the authors, who take full responsibility for the content of the article.

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